Level Up IAS

GS FOUNDATION 2024-25

GEOGRAPHY - 17

Geomorphic Processes - I

Inter Grav. Endo Exo (Exo of Earth) diastr (contin) catast Pan (Mount) Vola Earth Verti Horiz

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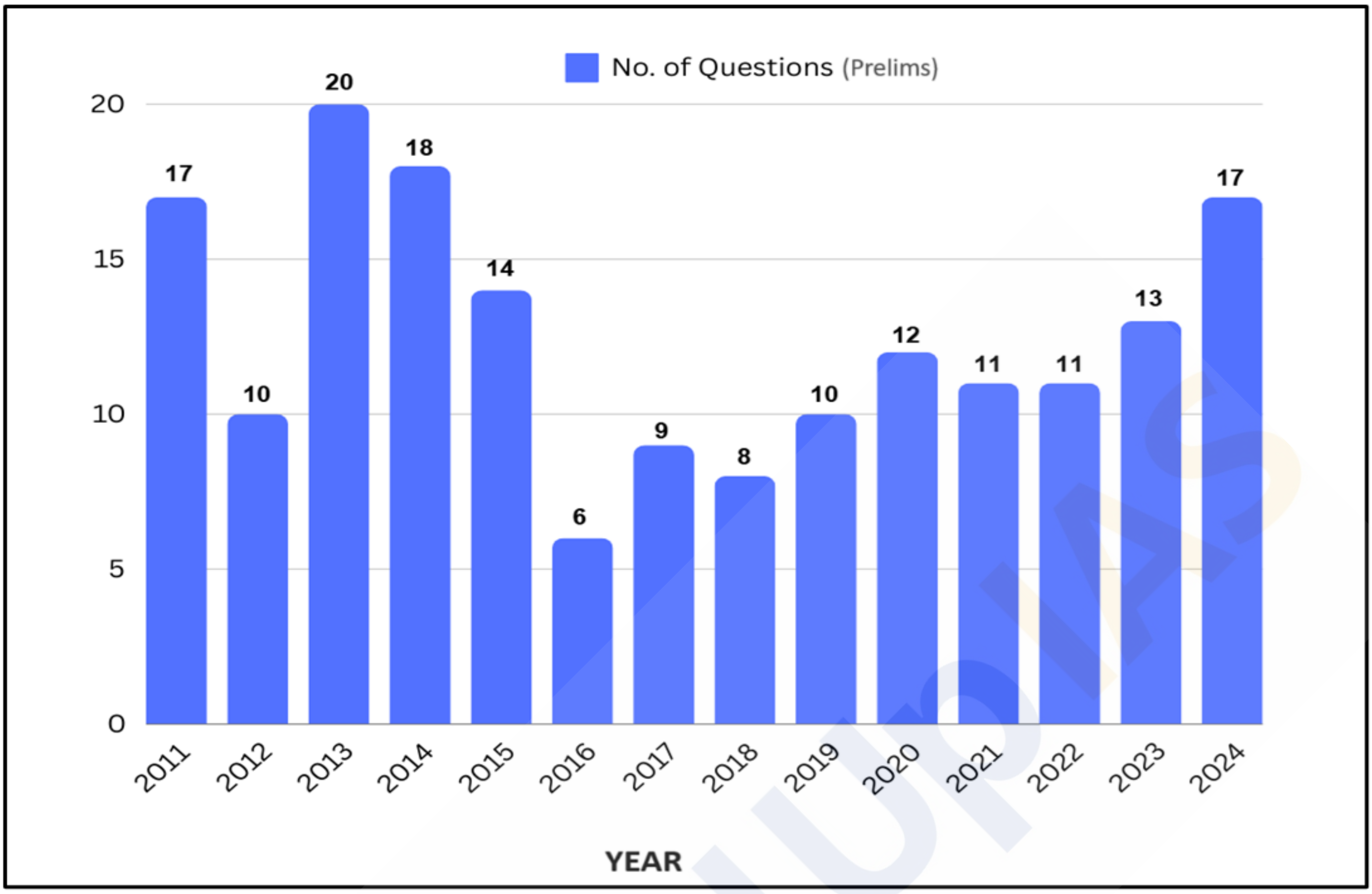

Topic-wise Trend Analysis (2011 – 2024)

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Geomorphology

1. Rock, Minerals, Rock cycle
2. Movement of Earth Crust → D,C,T

Rock cycle

3. CDT
SFS
CCC
PPT
} Theory on Earth Crust

4. Processes on Earth Surfaces

Geomorphic Process

→ Endogenic
→ Exogenic

Endogenic
→ Diastrophic
→ catastrophic

catastrophic
→ Volcano
→ Earthquake

Diastrophic
→ Vertical
→ Horizontal

Vertical
Epero.
→ continent building
→ Fault
→ Fold
→ wrapings
V.
D.

Horizontal
Orio.
↳ Mountain boulding
→ compression → Fold, Block → Reverse Fault → HW above
→ Divergence → Block → Normal Fault → FWall above
→ Transform
↳ Transform Fault → San Andres Fault

Geomorphic Processes

Introduction:

The Earth's surface is characterized by its unevenness, featuring diverse relief features like plains, plateaus, and mountains. This chapter aims to explore the reasons behind the **Earth's uneven terrain and the forces responsible** for shaping these remarkable relief features.

Geomorphic Processes:

The processes (Physical & Chemical) that bring about changes on the earth's surface are known as Geomorphic processes. These processes can be divided into two main categories: **Endogenic and Exogenic.**

Geomorphic Processes → change on Earth surface

- **Endogenic Processes** → Inside Energy, Interior
- **Exogenic Processes** → External Energy, Exterior

Endogenic Processes:

- The processes which ~~take place within the Earth~~ are energized from earth interior are called **Endogenic Processes.** The energy generating **from within the Earth** is the main force behind Endogenic Processes. These include:
  - Radioactivity → Disintegration of radio Active element in Earth Inside.
  - ~~Rotational Forces~~
  - ~~Tidal Friction~~
  - Primordial heat from the origin of the Earth.

> ***Do You Know?***
>
> **These forces slow down with time.**

- Endogenic Processes cause **two types of movements:**
  - **Horizontal Movements:** Side to side movements of the Earth's crust, causes lot of disruption as they involve **compression and tension** of the pre-existing rocks.
  - **Vertical Movements:** Originate from the center of the Earth causing large scale **upliftment or subsidence** of a part of the Earth's crust forming continents and plateaus.

Classification of Endogenic Processes:

- **Endogenic Processes**
  - **Slow Movements (*Diastrophism*)**
    - **Vertical (*Epeirogenic*)** continent building
      - Upward
      - Downward
    - **Horizontal (*Orogenic*)** Mountain Building
      - Compression
      - Tension
  - **Sudden Movements** Catastrophic Movement.
    - Volcanism
    - Earthquakes

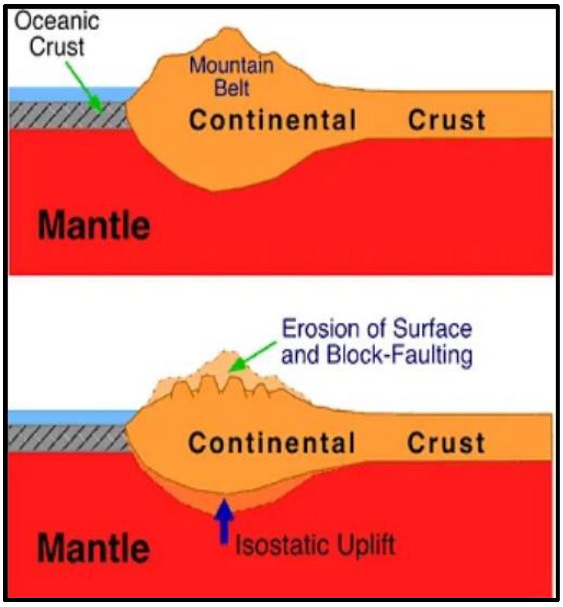
Plate Movement is diastrophic Endogenic Process

continent getting formed due to upward & Downward forces.

Diastrophism:

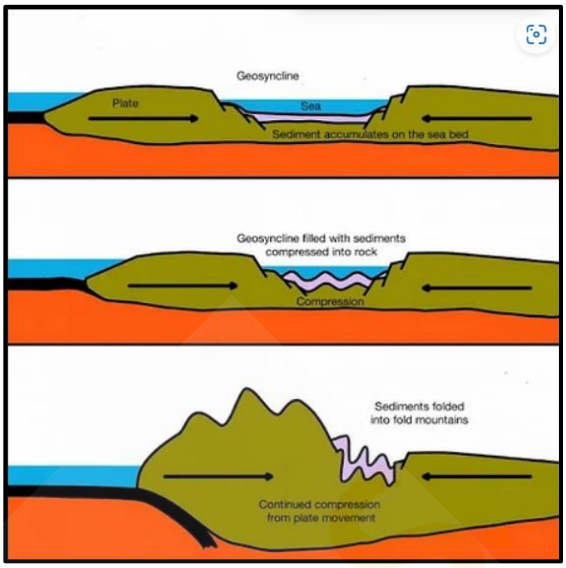
Diastrophism refers to all processes that **move, elevate, or deform the earth's crust** due to **diastrophic forces** which include **folding, faulting, warping and fracturing.** It can further be classified into **Epeirogenic movements** and **Orogenic movements.**

- **Epeirogenic or Continent building movements:** These are vertical movements of the Earth's crust that can cause the **uplift or subsidence** of large areas of land. It acts **along the radius** of the earth therefore; they are also called **radial movements.** Their direction may be towards (**subsidence**) or away (**uplift**) from the center.
  - **Uplift - Raised beaches, elevated wave-cut terraces, sea caves** and **fossiliferous beds** above sea level are evidences of uplift.
    - **Example:** Some of the beaches which have been elevated (uplifted) as much as 15 m to 30 m above the present sea level, such as **Kathiawar, Nellore,** and **Thirunelveli coasts.**
  - **Subsidence - Submerged forests, valleys** as well as buildings are evidence of subsidence.
    - **Example:** An earthquake (1819) submerged a part of **Rann of Kachchh.** Also, the Andamans and Nicobars have been isolated from the Arakan coast by the submergence of the intervening land.



Fold Mountain only
due to compression

- **Orogenic or Mountain building movements:** They are driven by the **interaction of tectonic plates (Earth's crust)** acting tangentially to the Earth's surface. It can further be divided into **Compressional** and **Tensional forces.**
    - **Compressional forces** - It results into folding, crustal bending, ~~local rise and subsidence because this type of force acts towards a point from one or more direction.~~
      fold Mountain
      Block Mountain
    - **Tensional forces** - It creates cracks, ruptures, features and faults, ~~since this type of forces acts away from a point in two directions.~~
      Block Mountain



Sudden Movements:

These are the result of **long period preparation deep within the Earth.** Only their cumulative effects on the Earth's surface are quick and sudden. It occurs mostly at the lithospheric plate margins. Geologically, these sudden forces are termed as **'constructive forces'** as it creates certain relief features on the Earth's surface. It can be of **two types: Earthquakes** and **Volcanoes.** (*** Already done in previous classes)

Structures produced by Endogenic Forces:

Endogenic forces create deformation in the Earth's crust and forms various structures as discussed below:

Folds:

Folds are bends in the Earth's crust owing to **compressional forces.** Folds can be formed by compressional forces, such as the collision of tectonic plates, or by extensional forces, such as the separation of tectonic plates.

Types of Folds:

- **Symmetrical Fold:**
    - These are ordinary folds. The limbs of the folds are equally inclined on either side. Their axial plane is vertical.

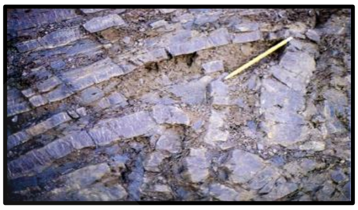
- **Asymmetrical Fold:**
    - One of the limbs is more inclined than the other. Their axial plane is inclined.

- **Monoclinal Fold:**
    - In this fold, one limb makes a right angle with the surface but the other limb is ordinarily inclined.

- **Isoclinal Fold:**
    - The two limbs are so much inclined in such a way that they appear equally inclined and parallel to each other.

- **Recumbent Fold:**
  - In this fold the two limbs are so much inclined that they become horizontal.

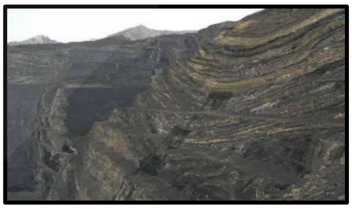
- **Overturned Fold:**
  - Here one limb is overturned over the other limb. The difference between the overturned and recumbent folds is that the overturned limbs are not horizontal like those of recumbent fold.



- **Plunge Fold:**
  - If the axis of the fold is not parallel to the horizontal but makes an angle with it, it is known as Plunging Fold.

- **Fan Fold:**
  - It is a great anticline which has many small anticlines and synclines. It is also known as Anticlinorium. A great syncline having many small anticlines and synclines is called Synclinorium.

- **Open Fold:**
  - If the angle between the limbs of a fold is obtuse, the fold is called Open fold.



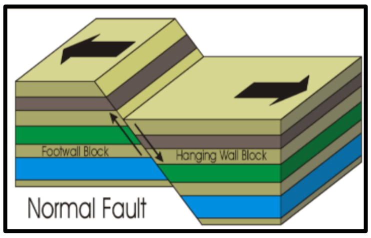
- **Closed Fold:**
  - If the angle between the limbs of a fold is acute, the fold is called Closed fold.

Faults:

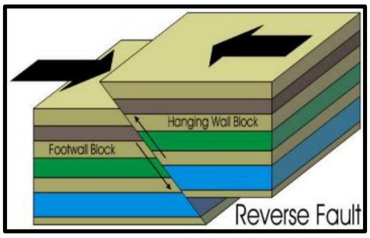
A fault is a fracture or zone of fractures in the Earth's crust where the **rocks on either side have slipped relative to each other.** This displacement occurs due to both **tensional and compressional** forces acting horizontally or vertically or both together at the same point of time.

Types of Faults:

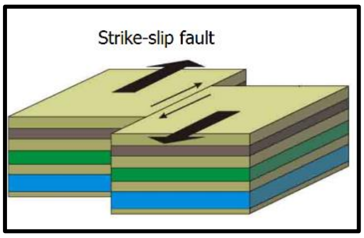
- **Normal Fault (Tension):**
  - Forms when **two blocks of the Earth's crust are pulled apart** due to **tensional forces.** The hanging wall (the block above the fault plane) moves down relative to the footwall (the block below the fault plane). It is generally associated with **divergent plate boundaries.**



- **Reverse Fault (Compression):**
  - Created by **compressional forces that push two blocks** of the Earth's crust together. In this case the hanging wall moves up relative to the footwall. It is often associated with convergent plate boundaries and **mountain-building processes.**



Strike-Slip Fault:



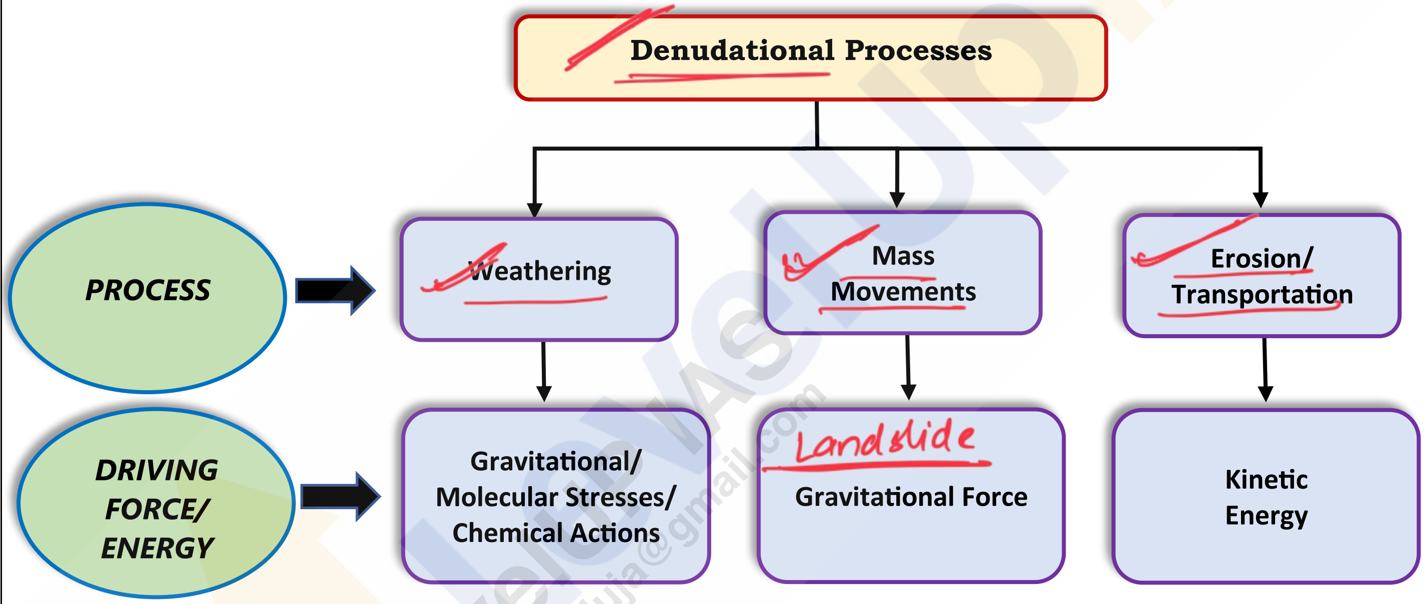
- Occurs due to **horizontal movement along the fault line** caused by **shearing forces.** The two blocks move past each other laterally in opposite directions. Common at transform plate boundaries like the **San Andreas Fault in California.**

Exogenic Processes:

Exogenic processes refer to the forces and processes that **operate on the Earth's surface,** responsible for reshaping the land and modifying its features through external mechanisms. It derives their energy from atmosphere determined by the **prime source, the Sun.**

These processes work on the Earth's surface and are primarily responsible for the **erosion, transportation,** and **deposition of materials. Temperature** and **precipitation** are the two important climatic elements that control various processes by inducing **stress** in earth materials.

All the exogenic processes are covered under general term **Denudation.** Denudation includes **weathering, mass wasting, erosion** and **transportation.**



For each process there exist a distinct driving force or energy. As there are different climatic regions on the Earth's surface the exogenic processes vary from region to region. The density, type and distribution of vegetation which largely depend upon precipitation and temperature exert influence indirectly on exogenic geomorphic processes.

Exogenic Processes also depend upon differences in wind velocities and directions, amount and kind of precipitation and its intensity, the relation between precipitation and evaporation, daily range of temperature, freezing and thawing frequency, depth of frost penetration etc. **The sole driving force behind all the exogenic process is the Sun.**

When climatic factors are common the **intensity of action depend on type and structure of rocks.** Structure includes folds, faults, orientation, inclination of beds, presence or absence of joints, bedding planes hardness, softness of constituent minerals, chemical susceptibility of mineral constituents, the permeability or impermeability.

Different types of rocks offer varying resistances to various geomorphic processes. **The effects of exogenic forces may be small and slow but in long run they have greater effects.** Hence the surface of the Earth is operated by different geomorphic processes and at varying rates.

Weathering:

Weathering is action of elements of weather and climate over Earth materials. Weathering is defined as **mechanical disintegration and chemical decomposition of rocks through the actions of various elements of weather and climate.** There are a number of processes within weathering which act either individually or together to affect the Earth materials in order to reduce them to fragmental state. Weathering processes are conditioned by many complex geological, climatic, topographic and vegetative factors.

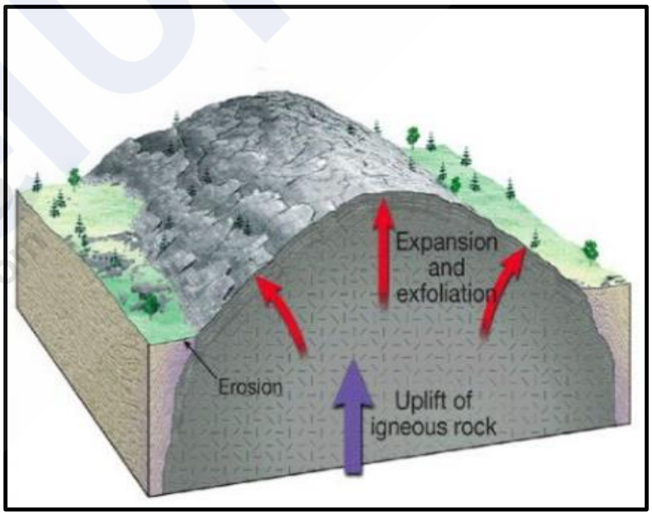
As very little or no motion of materials takes place in weathering. It is an **Insitu** disintegration or breakdown of rock material. It decreases the strength of the rocks but **increases permeability and porosity of the rock.**

There are three major groups of Weathering Processes:

1. **Physical or Mechanical Weathering.**
2. **Chemical Weathering.**
3. **Biological Weathering.**

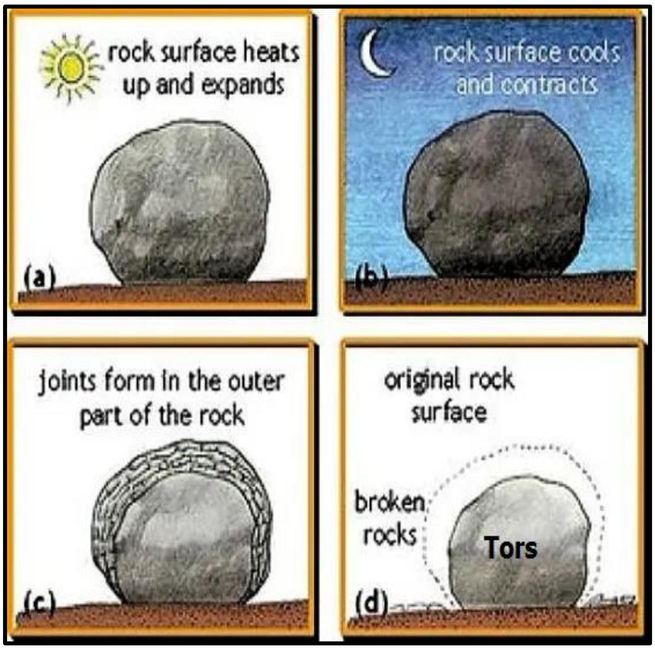
Physical or Mechanical Weathering:

Physical weathering, also known as **mechanical weathering,** is the process **of breaking down rocks and minerals into smaller pieces without changing their chemical composition.** This transformation of rocks into fragment particle occurs through mechanical methods such as **freezing, thawing, salt weathering, thermal cracking etc.** Physical weathering can be of several types:



Unloading and Expansion:

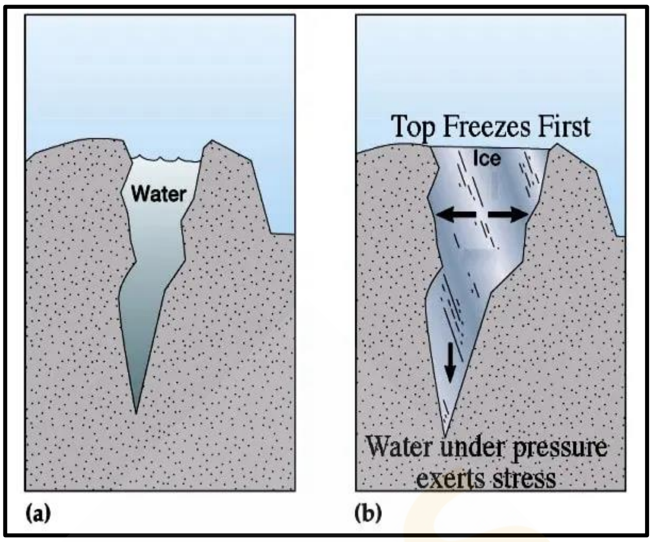
- Intrusive igneous rocks formed deep within the Earth's crust are subjected to immense pressure from the overlying layers. As **erosion gradually removes the top load, it leads to a release of vertical pressure.** Consequently, the uppermost layers of the **rock expand and develop fractures parallel to the surface.** Over time, these fractures cause sheets of rock to detach from the exposed surface, a process known as **"Exfoliation or Sheeting".**



Temperature Change and Expansion:

- Also known as **Thermal stress weathering.** Occurs when rocks are subjected to significant temperature fluctuations. It's particularly prominent in regions of **dry climates** and **high elevations** where diurnal (daily) or seasonal temperature variations are rapid. The process involves the **expansion and contraction** of rock materials due to temperature changes, resulting in stress within the rocks and thus ruptures the rock over the long term. It also creates exfoliation in rounded rocks. This exfoliation in rocks, like **granite,** creates round boulders with smooth surface called **'Tors'.**

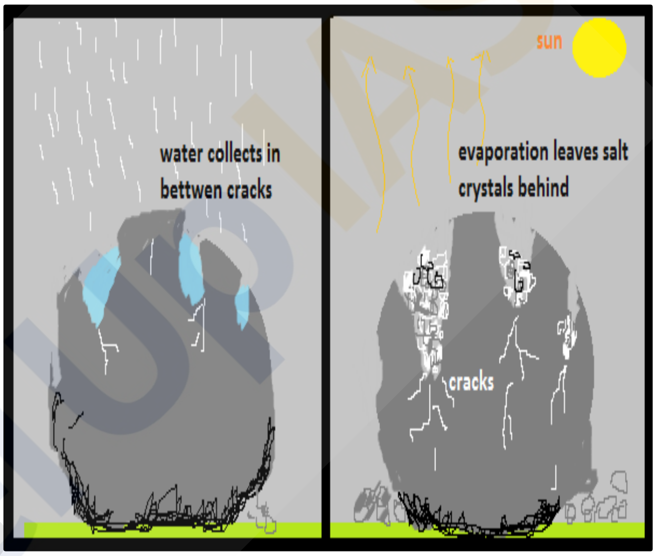
Freezing, Thawing and Frost Wedging:



- When water seeps into cracks or pores in rocks, **freezes, and expands during colder temperatures,** exerting pressure on the rock. These cycles of freezing and thawing cause the rock to weaken and break down over time, a process known as **'Frost wedging'**, contributing to the fragmentation of the rock into smaller pieces. It is found in **colder regions** where water freezes and thaws repeatedly.

> ***Do You Know?***
>
> ***The volume of water expands as much as by 9% when it freezes.***

Salt Weathering (Haloclasty):



- Salt weathering is most common in **coastal areas and in deserts,** where salt is abundant. It occurs when saline water infiltrate rock crevices or pores, subsequently crystallizing and expanding as the water evaporates. This expansion creates internal pressure, leading to the fracturing and breakdown of the rock. Salt weathering can also cause rocks to exfoliate.

Chemical Weathering:

Chemical weathering is the process by which rocks and minerals are broken down and decomposed by **chemical reactions.** It is caused by chemicals such as **water, acids, and oxygen.** Chemical weathering processes include **dissolution, solution, carbonation, hydration, oxidation** and **reduction** that act on the rocks to decompose, dissolve or reduce them to a fine state. Chemical weathering can be of several types:

Solution:

- Solution is the result of a **substance dissolving in water or acids,** creating a liquid mixture. The formation of a solution happens when solids dissolve due to **water or ion action** in water. It relies on the mineral's solubility in water or weak acids. Minerals like nitrates, sulphates, potassium, and others prone to dissolution are easily washed away in rainy climates and concentrate in arid regions. For instance, minerals such as **calcium carbonate and calcium magnesium bicarbonate in limestone dissolve in water with carbonic acid** and are carried away. Rock-forming minerals like common salt (sodium chloride) are similarly affected by this process of solution.

Carbonation:

$$CO_2 + H_2O \longrightarrow H_2CO_3$$
air, water, carbonic acid

$$CACO_3 + H_2CO_3 \longrightarrow Ca(HCO_3)_2$$
limestone, carbonic acid, calcium bicarbonate

- Carbonation involves the **reaction of carbonate and bicarbonate ions** with minerals, notably feldspars and carbonate minerals. This process is highly active in environments rich in carbon dioxide**. When carbon dioxide combines with water, it forms carbonic acid, a key factor in the breakdown of mineral surfaces.** The dissolution of calcium and magnesium carbonates in carbonic acid leads to the formation of caves, demonstrating the impact of carbonation. The presence of **atmospheric carbon dioxide, when dissolved in rainwater, induces acidity,** initiating the carbonation process.

Hydrolysis:

- Hydrolysis occurs when **mineral ions react with water ions (OH and H),** leading to the **decomposition of the rock surface.** New compounds are formed, increasing the solution's pH through the release of hydroxide ions. It is particularly efficient in weathering **common silicate and alumino-silicate minerals** due to their electrically charged crystal surfaces.

Oxidation and Reduction:

- Oxidation occurs between **minerals and oxygen leads to the formation of oxides and hydroxides.** It removes electrons from minerals, **making their structure less rigid and more unstable.** This reaction commonly results in red and yellow staining in soils, particularly in tropical regions **with high temperatures and rainfall, caused by iron and aluminium oxides.** Reduction occurs in environments lacking oxygen, typically below the water table or in waterlogged areas, causing the red colour of oxidized iron to turn greenish or bluish grey.

Chelation:

- In this process, the organic **chelating agent** which has **high affinity for metals, binds the metals** thus decomposing the rocks. (Humus breaking down to **humic acid** is example of chelation.)

Hydration:

- In hydration, **water molecules attach themselves firmly to minerals or compounds, resulting in an expansion of the mineral volume and rendering it less stable.** For instance, **calcium sulphate absorbs water and transforms into gypsum,** which is less stable. This process is **both reversible and gradual.** Repeated cycles of hydration can induce fatigue in rocks, potentially leading to their disintegration. The volume changes caused by hydration can also contribute to physical weathering through processes like **exfoliation and granular disintegration.** Furthermore, hydration can facilitate other decomposition reactions by expanding the crystal lattice, providing a larger surface area for further reactions to occur.

Do You Know?

These all-chemical weathering processes are inter-related to each other and go hand in hand and hasten the weathering process.

Biological Weathering:

It is the weakening and subsequent disintegration of rock by **plants, animals, and microbes.** It is contributed to or removal of ions and minerals from the weathering environment and physical variations due to movement or development of organisms. Key biological weathering processes include:

- **Root Wedging -** Plant roots growing into rock fractures can **exert pressure, causing the rock to crack and break apart.** As roots expand, they widen existing cracks, leading to rock disintegration.
- **Biological Activity -** Burrowing animals, such as **rodents and insects, contribute to weathering by creating tunnels and fissures in rocks,** which can lead to rock fragmentation and soil formation.
- **Lichen and Moss -** These organisms **secrete weak organic acids that break down minerals in the rock,** contributing to its disintegration. Their growth can also physically weaken the rock surface.
- **Decay -** The **decomposition of organic matter,** including dead plants and animals, **contributes organic acids** to the environment, accelerating the weathering of surrounding rocks.

Significance of Weathering:

- **Soil formation:** Weathering is essential for soil formation. As **rocks and minerals are weathered, they break down into smaller and smaller pieces,** which eventually form soil. Soil is important for plant growth and for supporting ecosystems.
- **Rock and mineral breakdown:** Weathering breaks down rocks and minerals into smaller pieces, making them easier to **transport by water, wind, and ice.** This process helps to create new landforms, such as mountains, valleys, and deltas.

Weather rock less surface Runoff more procolation

- **Groundwater recharge:** Weathering helps to **recharge groundwater supplies.** As rainwater infiltrates the ground, it dissolves minerals and other substances, which can then be taken up by plants or used by humans.
- **Landform Creation:** Weathering processes, alongside erosion, **play a pivotal role in the creation of various landforms,** including valleys, canyons, caves, and coastlines, contributing to the diversity of landscapes.
- **Contribution to the Carbon Cycle:** Chemical weathering **absorbs carbon dioxide** from the atmosphere, converting it into carbonate minerals, which are eventually deposited and stored in the form of sedimentary rocks.
- **Mineral release:** Weathering **releases minerals into the environment.** These minerals can be used by plants and animals, or they can be used to make products such as fertilizer and cement.
- **Economic resources:** Weathering can help to form valuable economic resources, such as **mineral deposits and bauxite** (the ore from which aluminium is extracted).
- **Environmental Balance:** Weathering processes are part of the **Earth's natural recycling mechanisms,** maintaining a balance in the Earth's geosphere, hydrosphere, and atmosphere.
- **Human Impact:** Understanding weathering processes is crucial in fields such as **agriculture, construction, and environmental conservation.** It influences soil quality, building materials, and landscape management.
- **Aids in Geological Studies:** By observing weathering patterns, geologists gain insights into **past environmental conditions,** helping in the interpretation of Earth's history.

Mass Movements:

Mass movement, often referred to as **Mass wasting,** is the process by which rock and debris are transferred down the slopes under the **direct influence of gravity.** This movement can occur at varying speeds, be it slow or fast, and it is primarily driven by the force of gravity. Unlike processes like erosion, which involve the actions of agents like water, glaciers, wind, or waves, mass movement is solely a product of gravity. **While weathering may aid in mass movements, it is not a prerequisite for them.**

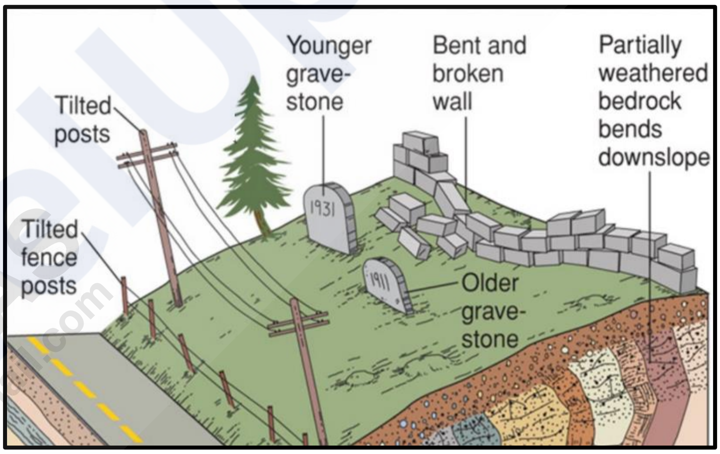
Key factors are responsible for Mass movements:

- **Gravity -** Primary force pulling materials downslope, more pronounced on steeper slopes.
- **Water -** Increases material weight and can act as a lubricant on slopes.
- **Rock and soil type -** Loose, unconsolidated materials are more prone to movement.
- **Vegetation -** Root systems help stabilize, while deforestation raises movement risk.
- **Human activity -** Construction, mining, and deforestation can disturb slopes, increasing vulnerability to mass movement.

Types of Mass movements:

- **Mass Movements**
  - Slow Movement
    - **Creep**
    - **Solifluction**
      ↳ soil saturated with water. so. it starts flowing
  - Rapid Movement
    - **Earthflow**
    - **Mudflow**
    - **Debris Avalanche**
  - Landslide
    - **Slump**
    - **Debris Slide**
    - **Rock Slide** Relatively slow to fall
    - **Rock Fall** very Rapid

Slow Movements:



Creep:

- Creep is the **slow, gradual, and imperceptible movement** of soil down a slope due to the **force of gravity.** This movement is often facilitated by processes such as **daily thawing and freezing or grazing, which loosen the soil and allow gravity to take effect.**
- **There are different types of creeps based on the material involved:**
  - **Soil Creep -** This involves **the slow movement of loose soil** under the influence of **gravity.**

  - **Talus Creep** - Also known as scree, it is the **gradual downslope movement of rock fragments.**

  - **Rock Creep -** This refers to the **intermittent slipping movement of rocks down a slope.**

Rapid Movements:

Rapid movements are most **common in humid climatic regions** and occur over a range of slopes, from gentle to steep. Various types of rapid movements include:

Earth Flow:

- Earth flow involves the movement of **water-saturated clayey or silty earth materials down low-angle terraces or hillsides.** On steeper slopes, even bedrock, particularly soft sedimentary rocks like shale or deeply weathered igneous rocks, may slide downslope.

Mudflow:

- Mudflows occur when thick layers of weathered materials, in the absence of vegetation cover and **under heavy rainfall, become saturated with water and flow down along definite channels.** This flow resembles a stream of mud within a valley. When mudflows emerge from channels onto piedmonts or plains, they can be highly destructive, engulfing roads, bridges, and houses. Mudflows **frequently happen on the slopes of erupting or recently erupted volcanoes.**

Debris Avalanche:

- Debris avalanches are more characteristic of humid regions, with or without vegetation cover, and **occur in narrow tracks on steep slopes.** These avalanches can be **much faster than mudflows** and are **similar to snow avalanches.**

Landslides:

- Landslides involve relatively **rapid and noticeable movements of dry materials.** The size and shape of the detached **mass depend on the nature of discontinuities in the rock, the degree of weathering, and the steepness of the slope.** Several types of movements are identified in this category:
    - **Slump -** The **slipping of one or several units of rock** debris with a **backward rotation** concerning the slope.
    - **Debris Slide -** Rapid rolling or sliding of earth debris **without backward rotation** of the mass.
    - **Debris Fall -** Nearly a **free fall of earth debris** from a vertical or overhanging face.
    - **Rockslide -** Sliding of individual rock masses **on bedding, joint, or fault surfaces.** Over steep slopes, rock sliding can be very fast and destructive, occurring as planar failures along discontinuities like steeply dipping bedding planes.
    - **Rockfall -** The free falling of rock blocks over any steep slope, **keeping themselves away from the slope.**

GS FOUNDATION 2024-25

GEOGRAPHY - 18

Geomorphic Processes-II

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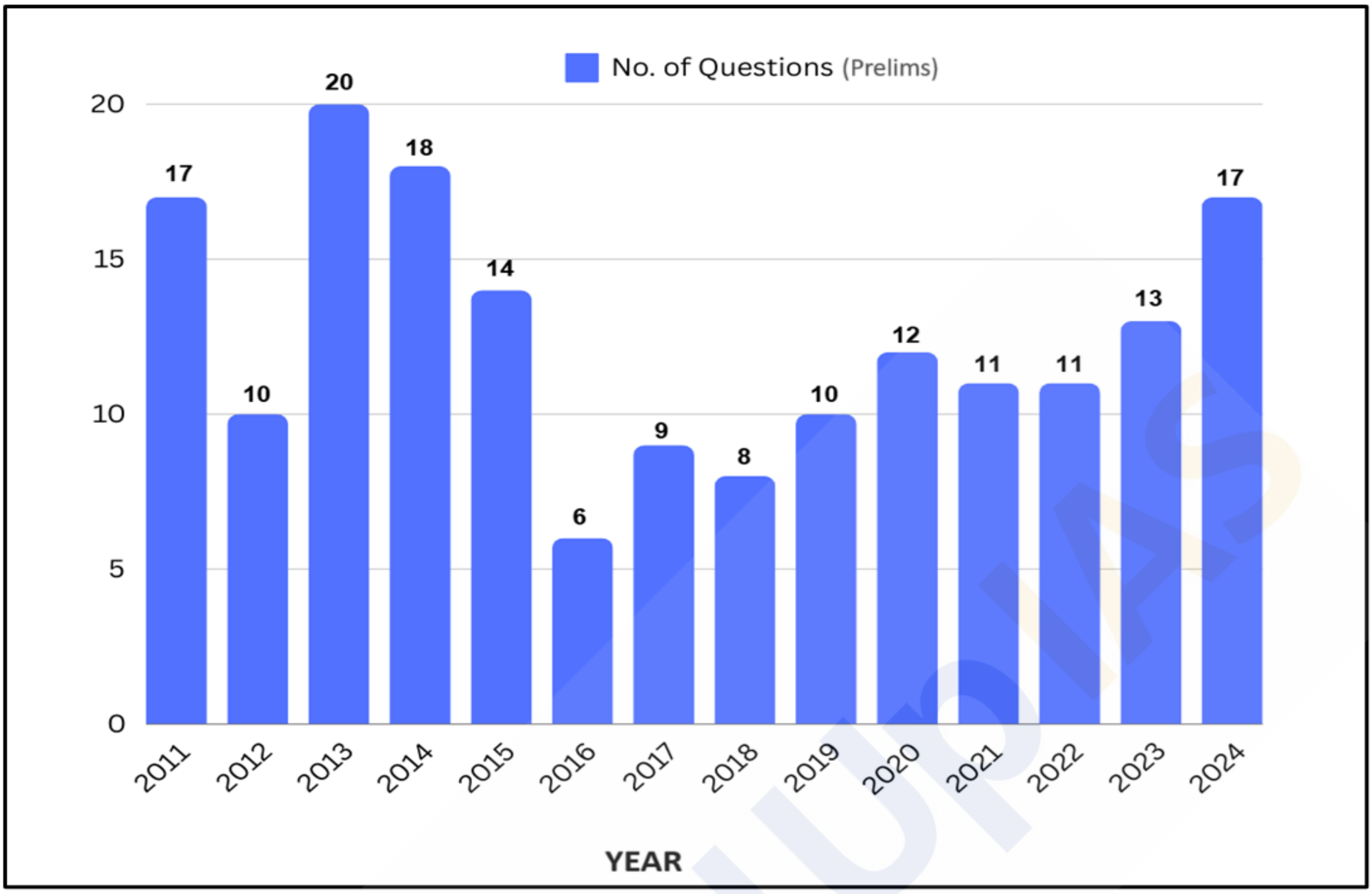

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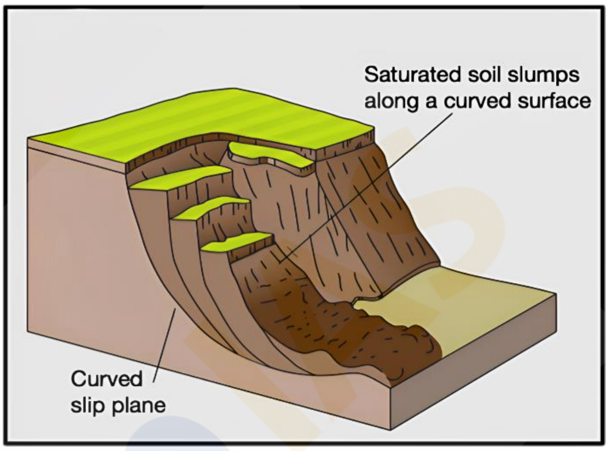
Landslides:

Landslides refer to the **downward movement of rock, soil, and other debris** on a slope due to gravity. They can be triggered by natural events such as **heavy rainfall, earthquakes, volcanic activity,** or human activities like **deforestation** and **construction.** They can occur suddenly and without warning, causing significant damage to property and loss of life. The type of landslide depends on several factors, including the type of material involved, the slope angle, the presence of water, and the triggering mechanism.

Types of Landslides:

Slump:

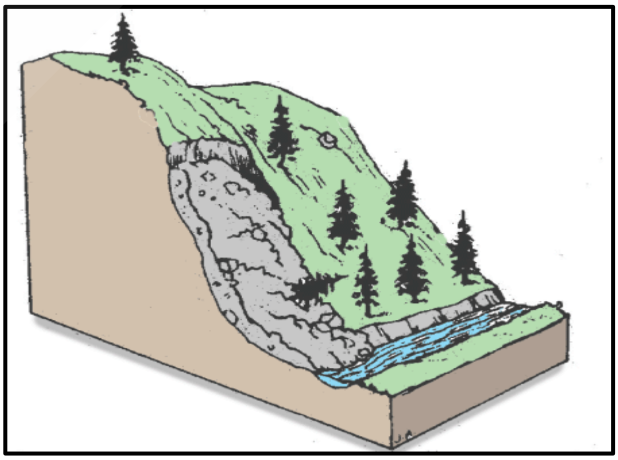


A slump is a type of landslide characterized by the **downward and outward movement (backward rotation) of a mass** of rock or unconsolidated material along a curved surface. This rotational movement typically results in a **concave-upward slip surface,** and the slumped material often forms a series of stepped terraces, or scarps.

Slumps are commonly triggered by the following factors:

- **Overloading -** Adding weight to the top of a slope, such as through **construction or the accumulation of water.**
- **Undercutting -** Removal of support at the base of a slope, often by natural processes like **river erosion** or human activities such as **mining.**
- **Water Saturation** - Increased pore **water pressure** reduces the friction between particles, making the slope more susceptible to failure.
- **Earthquakes -** Seismic activity can induce slumping by **shaking the ground** and weakening the slope structure.

Debris Slide:

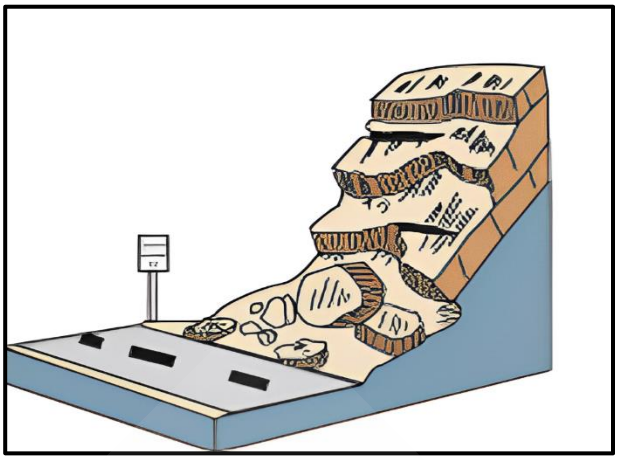


A debris slide involves the **rapid downslope movement (without backward rotation) of unconsolidated soil, rock, and organic material.** Unlike slumps, debris slides **do not exhibit rotational movement** but instead slide along a more or less planar surface.

Key characteristics and triggers include:

- **Loose Material -** Debris slides typically involve a mixture of **soil, rock, and vegetation.**
- **Steep Slopes -** Occur on steep inclines where **gravitational forces are significant.**
- **Rainfall -** Heavy or prolonged rainfall can saturate the material, **reducing cohesion** and increasing the likelihood of sliding.
- **Disturbance -** Natural events like **earthquakes or human activities** such as deforestation and construction can destabilize the slope.

Debris Fall:

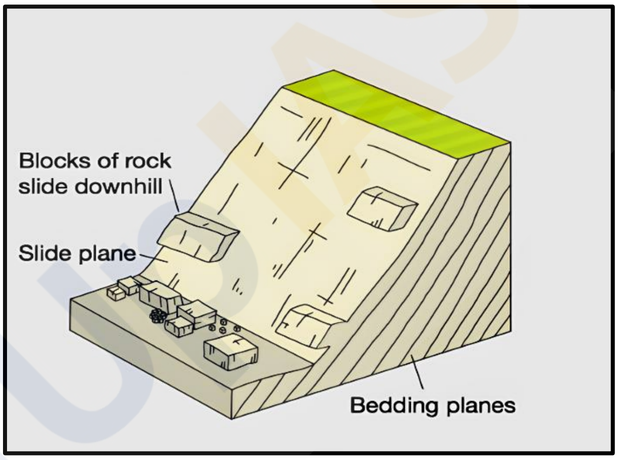


A debris fall is a **sudden, free-fall movement** of rock, soil, and debris from a steep slope or cliff.

This type of landslide is characterized by:

- **Free-Fall Motion -** Material falls freely through the air and accumulates at the base of the slope.
- **Steep Terrain -** Typically occurs on **vertical or near-vertical slopes.**
- **Triggering Events -** Earthquakes, weathering, and freeze-thaw cycles can dislodge material.
- **Accumulation Zone -** The fallen debris forms a talus or scree slope at the base of the cliff.

Rockslide:



A rockslide involves **the rapid downslope movement of large blocks of bedrock.**

This type of landslide is distinguished by:

- **Block Movement -** Large, coherent blocks of **rock slide along a defined planar surface or series of surfaces.**
- **Structural Weaknesses - Joints, faults, and bedding** planes within the rock provide pathways for sliding.
- **Triggers - Earthquakes, weathering, and the removal of support at the base** of the slope can initiate rockslides.
- **High Velocity -** Rockslides can move at **high speeds** and travel long distances, posing significant hazards.

Factors Affecting Landslides:

Geological Factors:

- **Rock Type -** Soft or fractured rocks are more prone to landslides.
- **Structural Features -** Faults, joints, and bedding planes can act as slip surfaces.

Topographical Factors:

- **Slope Angle -** Steeper slopes are more susceptible to landslides.
- **Elevation and Relief -** Higher elevations and significant relief differences can increase the potential for landslides.

Hydrological Factors:

- **Rainfall -** Intense or prolonged rainfall can saturate the ground and trigger landslides.
- **Groundwater -** High groundwater levels reduce slope stability. Ground water near to Water surface more chance of solification.
  Eg. Bihar, 1934

Vegetation:

- **Root Systems -** Vegetation can stabilize slopes through root reinforcement.
- **Deforestation -** Removal of vegetation reduces slope stability and increases landslide risk.

Human Activities:

- **Construction -** Excavation and construction can destabilize slopes.
- **Mining and Quarrying -** Extractive activities can remove support from slopes.
- **Land Use Changes -** Changes in land use, such as urbanization, can increase the risk of landslides.

Landslide Prevention and Mitigation:

Slope Stabilization:

- **Retaining Walls -** Constructing retaining walls to support and stabilize slopes.
- **Terracing -** Creating terraces to reduce slope angles and minimize erosion.

Drainage Control:

- **Surface Drains -** Installing surface drains to divert water away from slopes.
- **Subsurface Drains -** Installing subsurface drains to reduce groundwater pressure.

Reforestation:

- **Planting Vegetation -** Establishing deep-rooted plants to reinforce the soil.
- **Bioengineering -** Using plants and natural materials to stabilize slopes.

Land Use Planning:

- **Zoning Regulations -** Implementing zoning regulations to restrict development in high-risk areas.
- **Risk Mapping -** Creating landslide hazard maps to guide land use planning and development.

Monitoring and Early Warning Systems:

- **Instrumental Monitoring -** Using sensors and instruments to monitor slope stability and detect early signs of movement.
- **Early Warning Systems -** Developing early warning systems to alert communities of impending landslides.

Engineering Solutions:

- **Rock Bolts and Anchors -** Installing rock bolts and anchors to secure unstable rock masses.
- **Grouting -** Injecting grout into fractures and voids to strengthen the slope material.

Debris Avalanche and Landslide in India:

Debris avalanches and landslides are significant natural hazards in various parts of India, particularly in the Himalayas and the Western Ghats. These events can have devastating **impacts on the environment, infrastructure, and human lives.** The causes and characteristics of these hazards differ between the two regions due to their distinct geological and climatic conditions.

In the Himalayas:

The Himalayas are one of the youngest and most dynamic mountain ranges in the world. They are characterized by high seismic activity, steep slopes, and a complex geological structure. Several factors contribute to the frequent occurrence of debris avalanches and landslides in this region:

- **Tectonic Activity:**
    - The Himalayas are **tectonically active** due to the ongoing collision between the Indian Plate and the Eurasian Plate. This tectonic activity leads to **frequent earthquakes, which can trigger landslides** and avalanches by destabilizing slopes.

- **Geological Composition:**
    - The Himalayas are primarily composed of **sedimentary rocks and unconsolidated or semi-consolidated deposits.** These materials are **inherently less stable than igneous or metamorphic rocks,** making the region more susceptible to slope failures.
- **Steep Slopes:**
    - The slopes in the Himalayan region are extremely steep. The combination of **steep gradients** and loose geological materials significantly increases the risk of landslides and debris avalanches.
- **Weathering and Erosion:**
    - Intense weathering and erosion processes, driven by **heavy rainfall, snowmelt, and temperature variations,** further weaken the slopes, contributing to the instability.

In the Western Ghats:

The Western Ghats and the **Nilgiris,** which border Tamil Nadu, Karnataka, and Kerala, are relatively tectonically stable compared to the Himalayas. However, debris avalanches and landslides still occur in this region due to several factors:

- **Tectonic Stability and Geological Composition:**
    - Unlike the Himalayas, the Western Ghats are **mostly made up of very hard, crystalline rocks.** Despite being tectonically stable, the region experiences landslides due to other geomorphological and climatic factors.
- **Steep Slopes and Escarpments:**
    - Many slopes in the Western Ghats and Nilgiris are steep, with **almost vertical cliffs and escarpments.** The steep terrain enhances the gravitational force acting on the slopes, making them prone to landslides.
- **Mechanical Weathering:**
    - Mechanical weathering, driven by **temperature fluctuations,** is significant in the Western Ghats. The **expansion and contraction of rocks** due to temperature changes lead to the development of cracks and fractures, weakening the rock structure over time.
- **Heavy Rainfall:**
    - The Western Ghats receive **heavy rainfall, particularly during the monsoon season.** The intense and prolonged rainfall saturates the soil and rock, reducing their shear strength and leading to slope failures. The high rainfall also contributes to the formation of landslide-prone debris flows.

Erosion:

Erosion is a natural process involving the **detachment and removal of loosened rock materials and soils by external processes,** free from human interference, and is also known as **geological erosion.** The slow removal of soil is a fundamental aspect of the geological process of **denudation,** which is both inevitable and universal.

Accelerated erosion refers to the **increased rate of erosion caused by human-induced land-use changes.** Therefore, when we refer to soil erosion, we usually mean accelerated erosion, also known as **man-induced erosion.**

Soil Erosion:

Soil erosion involves the **loosening and displacement of topsoil particles.** This can occur as a slow, natural process (geological erosion) or as a rapid process driven by **deforestation, floods, tornadoes,** or other **human activities.** Soil erosion represents an **extreme form of soil degradation,** where the natural geomorphological process is accelerated to such an extent that soil is removed at a rate ten to several thousand times faster than under conditions of natural vegetation, far outpacing the rate of new soil formation.

Accelerated soil erosion, or man-induced soil erosion, is particularly **prevalent in humid climate regions where extensive forest clearance, grassland removal, and overgrazing and trampling by livestock** have occurred at alarming rates due to human activity.

Soil erosion primarily involves two processes:

1. **Loosening and detachment of soil particles** from the soil mass
2. **Removal and transport of the detached soil** particles downslope

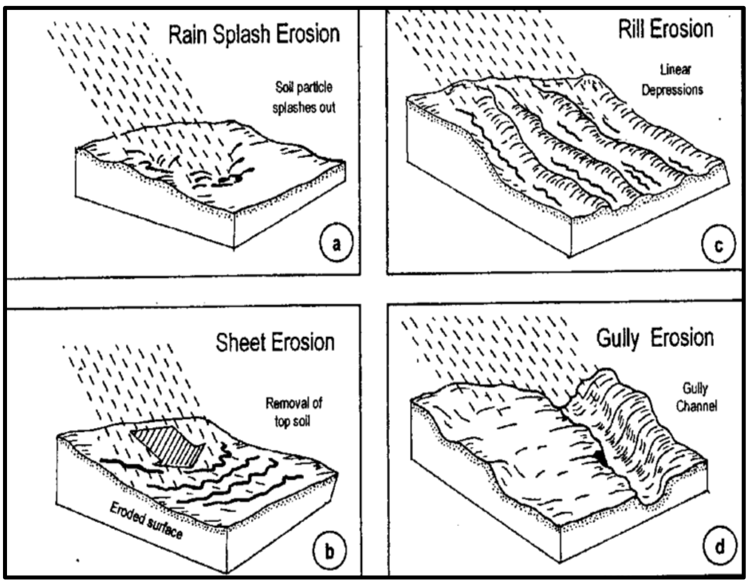
Soil erosion and land degradation together, constitute one of the major problems that disturb the ecological balance of the world. The rapid increase in the human population has placed a great strain on the land and soil resources resulting in land degradation and soil erosion. On a worldwide basis, more than 4.85 billion acres (1.96 billion hectares) or **17% of the earth under vegetation has been degraded by humans to various extents.**

Types of Soil Erosion:

Water Erosion:

This type of erosion is caused by the **action of water, which can remove soil** through the impact of raindrops and the movement of surface water.

Depending on how the soil is lost, water erosion can be categorized as:



- **Splash Erosion -** This is the initial stage of the erosion process. It occurs when **raindrops hit bare soil with enough force to break up soil aggregates.** The explosive impact disperses individual soil particles, causing them to 'splash' onto the soil surface.
- **Sheet Erosion -** In this form, **a thin layer of soil is uniformly removed from a large area.** This type of erosion occurs when water flows over the soil surface in a dispersed manner, carrying away a consistent layer of soil.

- **Rill Erosion -** When sheet erosion progresses with greater intensity**, the runoff water flows rapidly over the soil surface, carving out well-defined, finger-shaped grooves or channels.** These small channels are known as rills and indicate a more advanced stage of erosion compared to sheet erosion.
- **Gully Erosion -** Gully erosion occurs when several rills converge, usually on a steep slope. This convergence forms wider and deeper channels called gullies, which can significantly alter the landscape.

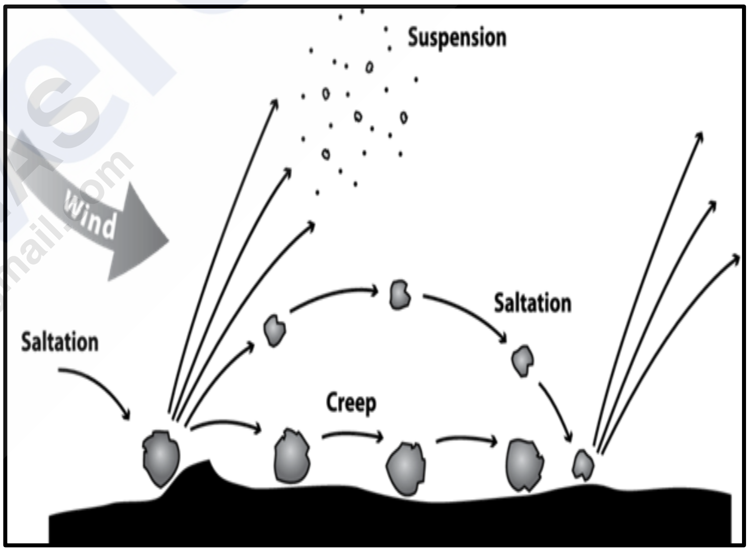
Methods to control Water Erosion:

- **Contour Bunding - Building embankments along the contours** of the land to slow down water runoff and encourage water infiltration.
- **Channeling -** Constructing **channels to direct water flow and reduce erosion.**
- **Gully Path Modifications -** Altering the paths of gullies to control water flow and prevent further erosion.
- **Changing Cropping Patterns -** Implementing **crop rotation and other agricultural practices** to maintain soil cover and reduce erosion.
- **Land Leveling - Smoothing out the land surface** to reduce runoff and encourage water absorption.

Wind Erosion:

Wind is a formidable agent of erosion, driving aeolian processes that **transport dust, sand, and ash.** Wind erosion is **prevalent in arid regions where the soil is primarily sandy, and vegetation is sparse or non-existent.** When strong winds blow over exposed topsoil, it can be carried away in dust or sand storms.

There are three main types of wind erosion:



- **Saltation -** This is the primary method of soil movement by wind. In saltation, **fine soil particles are lifted into the air and drift horizontally across the surface.** These particles travel much farther horizontally than they do vertically, typically about four times as far.
- **Suspension -** Here, the smallest **soil particles are carried into the air by the wind, forming a fine dust that can be transported over long distances.** These particles remain suspended in the air, moving with the wind over greater distances.
- **Surface Creep -** Heavier **soil particles that are too large to be lifted into the air are pushed along the surface by the wind.** This process involves the gradual movement of these larger particles across the ground.

Methods to control Wind Erosion:

- **Vegetative Cover - Planting grasses, shrubs, or trees** can help stabilize the soil and reduce wind speed near the surface. Vegetative cover **acts as a barrier to the wind** and helps bind the soil with root systems.
- **Windbreaks and Shelterbelts -** Establishing rows of trees or shrubs, known as windbreaks or shelterbelts, can reduce wind speed and protect soil. These structures can be **placed around fields or along the edges of farmland.**

- **Soil Moisture Management -** Keeping the soil moist can help reduce wind erosion since wet soil is less likely to be blown away compared to dry, loose soil. Techniques like **mulching or using irrigation systems** can help maintain soil moisture.
- **Tillage Practices -** Adopting conservation tillage practices, such as **reduced tillage or no-till farming,** helps maintain soil structure and reduces the amount of loose soil available for wind to carry away.
- **Erosion Control Mats and Mulches -** Applying materials like erosion control mats or mulches can protect soil surfaces from the wind. These materials can be made from natural or synthetic fibers and help to stabilize the soil and reduce erosion.

Glacial Erosion:

Glacial erosion refers to the process by which **glaciers shape and modify the Earth's surface through the removal and transportation of sediment and rocks.** It is influenced by various factors such as **basal sliding, tectonic rock uplift,** and **climate change.** In cold regions and high mountains, glaciers move slowly, transporting everything from sand to boulders.

As glaciers move, they grind against the ground, eroding both the earth and the rocks they carry. This process creates basins and **steep-sided valleys,** leaving behind visible sediment called moraine.

Ice Age glaciers shaped many landscapes, including the **Finger Lakes in New York and the fjords of Scandinavia.** Today, glaciers in **Greenland** and **Antarctica** continue to erode the earth, although measuring this erosion is challenging due to the ice sheets' thickness. These ice sheets erode rapidly, up to half a centimeter (0.2 inches) annually.

Factors affecting Soil Erosion:

- **Climate -** Weather conditions, including rainfall intensity and frequency, play a crucial role in erosion. Heavy rainfall can lead to increased surface runoff, which can cause soil detachment and transport. Conversely, dry climates may experience less erosion but can still be affected by wind erosion.
- **Soil Type -** Different soil types have varying susceptibilities to erosion. Sandy soils are generally more prone to erosion due to their loose structure, while clay soils are more cohesive and less easily eroded. Soil texture, structure, and organic matter content all influence erosion rates.
- **Vegetation Cover -** Vegetation acts as a natural barrier to erosion by stabilizing soil with root systems and reducing surface runoff. Areas with dense vegetation experience less erosion compared to bare or sparsely vegetated areas.
- **Topography -** The slope and shape of the land surface affect erosion. Steeper slopes tend to have higher erosion rates due to increased runoff velocity, which can carry away more soil. The shape of the land can also influence how water and wind move across it.
- **Land Use -** Human activities such as deforestation, agriculture, and construction can significantly impact erosion. Practices like ploughing, logging, and urban development can remove vegetation, disturb soil, and increase erosion rates.
- **Water Flow -** The presence and flow of water, including rivers, streams, and rainfall runoff, are major erosion factors. The speed and volume of water flow can erode riverbanks, create gullies, and transport sediment.
- **Wind -** In arid and semi-arid regions, wind can be a significant factor in erosion. Strong winds can lift and carry away loose, dry soil particles, leading to desertification and land degradation.

- **Soil Conservation Practices -** Measures such as terracing, contour ploughing, and cover cropping can help reduce erosion by managing water flow and stabilizing soil. Effective soil conservation practices can significantly mitigate erosion.
- **Geological Factors -** The underlying geology of an area, including rock type and structure, influences erosion. Softer, more erodible rocks and sediments are more easily worn away compared to harder, more resistant materials.
- **Human Activity -** Construction, mining, and other land-disturbing activities can exacerbate erosion. Alterations to the landscape, such as removing vegetation or changing natural water flow patterns, can increase erosion rates.

Positive Impacts of Soil Erosion:

- **Soil Formation -** Erosion helps in the formation of new soil layers. As materials are broken down and transported, they contribute to the development of fertile soil in other locations.
- **Nutrient Redistribution -** Erosion can move nutrient-rich sediments from one area to another, potentially enhancing soil fertility in regions where nutrients are needed.
- **Creation of Landforms -** Erosion shapes landscapes, creating various landforms such as valleys, canyons, and coastal cliffs, which contribute to natural beauty and ecological diversity.
- **Sediment Deposition -** Eroded materials often settle in new areas, forming river deltas and floodplains that can be rich in nutrients and support agriculture.
- **Aquifer Recharge -** In some cases, eroded materials can help in the recharge of aquifers by increasing the porosity of soil and allowing water to infiltrate deeper into the ground.
- **Habitat Creation -** Erosion can create diverse habitats for plants and animals, such as rocky outcrops and sediment-rich wetlands.
- **Improved Water Flow -** Erosion can sometimes lead to the formation of channels and pathways that help in managing and directing water flow, reducing flood risks.
- **Increased Biodiversity -** The changing landscape due to erosion can create niches for different species, enhancing local biodiversity.
- **Sediment Supply to Ecosystems -** Erosion provides a continuous supply of sediments to coastal and riverine ecosystems, which can be crucial for maintaining certain ecological functions.
- **Cultural and Scientific Value -** Erosion exposes geological layers and fossils, offering valuable insights into Earth's history and contributing to scientific research.

Negative Impacts of Soil Erosion:

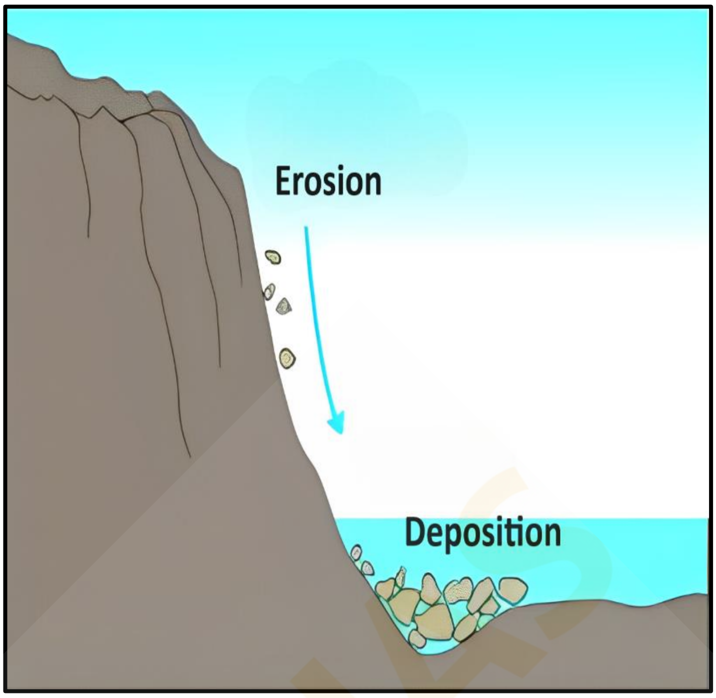
- **Soil Loss -** Erosion leads to the loss of fertile topsoil, which can severely impact agricultural productivity and food security.
- **Decreased Land Productivity -** As soil is eroded, the remaining soil may become less productive, leading to lower crop yields and degraded farmland.
- **Water Quality Degradation -** Eroded sediments can carry pollutants into water bodies, causing water quality issues and harming aquatic life.
- **Infrastructure Damage -** Erosion can undermine roads, buildings, and other infrastructure, leading to costly repairs and potential safety hazards.

- **Increased Flood Risk -** The loss of vegetation and soil structure due to erosion can increase the risk of flooding by reducing the land's ability to absorb and retain water.
- **Loss of Agricultural Land -** Persistent erosion can lead to the loss of large areas of agricultural land, impacting local economies and food production.
- **Ecosystem Disruption -** Erosion can disrupt ecosystems by altering habitats and leading to the loss of plant and animal species that depend on stable environments.
- **Sedimentation in Waterways -** Excessive sedimentation in rivers and streams can affect water flow, reduce aquatic habitats, and increase the risk of reservoir siltation.
- **Increased Maintenance Costs -** Areas affected by erosion often require increased maintenance and conservation efforts to prevent further degradation and mitigate its impacts.
- **Economic Losses -** The overall economic impact of erosion, including reduced land value, increased maintenance costs, and loss of productivity, can be significant for communities and economies.

Methods of Controlling Soil Erosion:

- **Vegetative Cover -** Planting grass, shrubs, or trees helps stabilize the soil with their root systems. Vegetation acts as a protective layer that reduces the impact of rainfall and slows down surface water flow, preventing soil particles from being washed away.
- **Contour Plowing -** This agricultural practice involves plowing along the contour lines of a slope rather than up and down. It helps to create natural barriers that slow down water runoff and encourage water to infiltrate into the soil, reducing erosion.
- **Terracing -** Building terraces on steep slopes creates flat areas that slow water runoff and capture soil. This method effectively reduces the speed of water flow and provides areas for vegetation to grow, further preventing erosion.
- **Cover Crops -** Growing cover crops like clover or rye during the off-season helps protect the soil from erosion. These crops keep the soil covered, reduce runoff, and their root systems help bind the soil together.
- **Reforestation and Afforestation -** Planting trees in deforested or barren areas helps to stabilize the soil. Tree roots improve soil structure and reduce the speed of surface water runoff, while tree canopies protect the soil from direct rainfall.
- **Gully Plugging -** Filling in and stabilizing eroded gullies with materials like rocks, vegetation, or constructed barriers helps prevent further erosion. Gully plugging slows water flow and encourages sediment deposition.
- **Erosion Control Blankets -** These are mats made from natural or synthetic fibers that are placed on bare soil. They protect the soil from wind and water erosion while helping to establish vegetation.
- **Building Retaining Walls -** Constructing walls on steep slopes can help prevent soil from sliding or washing away. Retaining walls can be made from various materials like stone, concrete, or timber and are often used in urban and agricultural settings.
- **Proper Drainage Systems -** Installing and maintaining effective drainage systems helps manage water flow and prevent excessive runoff. Techniques such as installing drainage ditches, French drains, or rain gardens can help direct and manage water to prevent erosion.
- **Reduced Tillage -** Minimizing tillage or practicing no-till farming helps to maintain soil structure and organic matter. Reduced tillage leaves crop residues on the soil surface, which protects the soil from erosion and helps retain moisture.

Deposition:



Deposition, a **direct consequence of erosion,** plays a vital role in shaping the Earth's surface. As erosional agents such as running water, glaciers, wind, waves, and groundwater move across the landscape, they eventually lose their velocity and energy, especially when encountering gentler slopes. This reduction in energy causes the materials they carry to settle.

Deposition is a **passive process, occurring naturally as these transported materials are laid down.** The sequence of deposition typically begins with **coarser materials settling first, followed by finer ones.** This process often fills depressions created by previous erosional activities, contributing to the gradual levelling of the terrain.

Thus, the **same forces responsible for erosion also act as agents of deposition,** continually modifying and reshaping the Earth's surface.

===== XXXXX =====

Land slide

Normal Reason

(1) Cloud Burst
↳ Solification
↳ less perocation
↳ more surface Runoff

(2) EQ → CC

(3) soft sedimentory rock
↳ landslide

(4)

Human Reason

(1) Dam → Additional load on rock.

(2) construction

(3) Mining

(4) Deforestation
↳ lack of root

(5) Excess Tourist pressure beyond balance.

(6) GLOF
↳ Glacier Lake flood outburst
↳ Sikkim

ə)

Glacier upward { sand } do later }

GS FOUNDATION 2024-25

GEOGRAPHY - 19

Lecture 21

Landforms and Their Evolution

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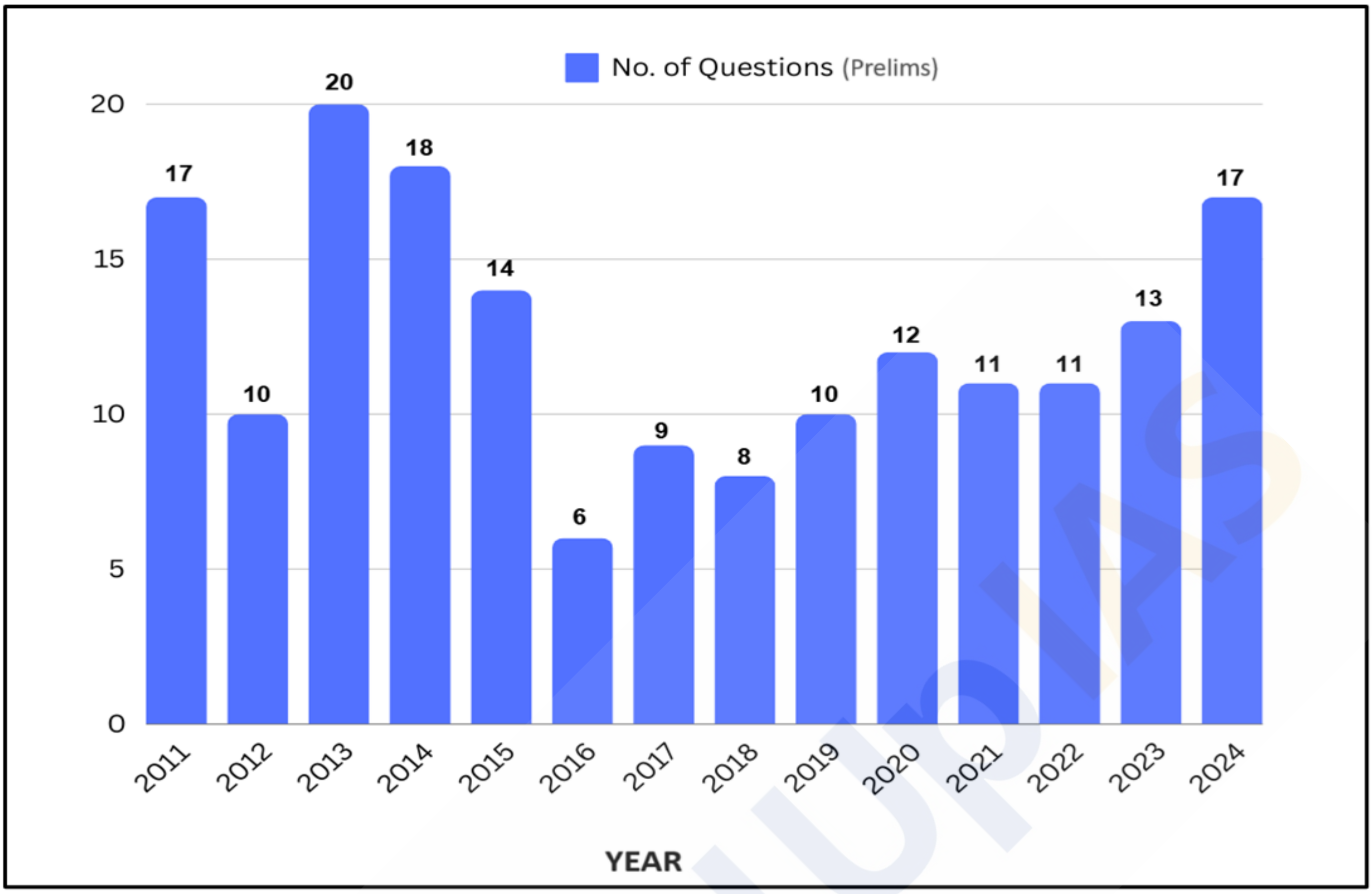

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Landforms and Their Evolution

Introduction:

After weathering processes act on the materials that make up the earth's surface, geomorphic agents such as **running water, groundwater, wind, glaciers,** and **waves** take over to perform erosion. It is well known that erosion causes significant changes to the earth's surface. **Following erosion, deposition occurs, which also leads to changes on the surface of the earth.**

Landforms are small to medium-sized areas of the earth's surface. When several related landforms come together, they create landscapes, which are larger tracts of the earth's surface. Each landform has a **distinct physical shape, size, and composition,** and these characteristics result from specific geomorphic processes and agents. Most of these processes and agents work slowly, so it takes a long time for their effects to become noticeable. Every landform has an origin and evolves over time.

Once a landform is created, it can **change in shape, size, and nature either gradually or rapidly,** depending on the continued action of geomorphic processes and agents. Changes in climatic conditions and the vertical or horizontal movements of land masses can alter the intensity or nature of these processes, leading to further modifications in landforms.

Evolution in this context refers to the **stages of transformation** that parts of the earth's surface undergo, either changing from one landform to another or altering individual landforms after their initial formation. Each landform has a history of development and changes over time, similar to the stages of life—**youth, maturity,** and **old age.**

Landforms Made by Running Water or Fluvial Landforms:

Fluvial landforms are **those landforms which are shaped and modified by the running water.** The term "fluvial" is derived from the Latin word **"fluvius,"** meaning **'river.'** Running water has sculpted most of the land surface across the world in comparison to other agents of erosion like wind or Glacier. Fluvial landforms exhibit a wide range of dimensions, from small features like rills to large-scale units such as major rivers and their extensive drainage basins. Remarkably, **rivers flowing to the oceans drain about 68% of the Earth's land surface.**

Rivers typically originate in upland regions where the slope facilitates runoff. These upland areas serve as the catchment zones for rivers, with the **crest of mountains forming divides or watersheds.** From these divides, streams flow down the slopes. The initial stream that forms due to the slope is known as a **'consequent stream.'** As this stream erodes the surface, it is joined by numerous tributaries from either side.

A **drainage basin,** or **watershed,** is a **fundamental landscape unit in fluvial geomorphology.** It includes a primary, or trunk, river and its tributaries. The drainage basin is crucial for understanding how water and sediments move through a landscape. Erosion → At start Deposition ↳ At End

Erosion by Running Water occurs through various processes:

- **Hydration -** The **physical force of running** water wearing down rocks.
- **Corrosion or Solution - Chemical weathering** caused by the interaction of water with minerals in rocks.
- **Attrition -** When the **stream loads** (pebbles, sand, silt etc) **move together, they cause their own wear and tear** by colliding with one another. This rounding and shaping of these pebbles and boulders amongst themselves is called attrition.
- **Corrasion or Abrasion - Solid particles** carried by the river **strike against rock surfaces,** wearing them down. Scrapping the surface
- **Downcutting (Vertical Erosion) -** The **deepening of a stream** channel. → Generally at Mountain top V. shaped formation.

River capture is generally through headward erosion → Eg. Ganga dyamuna River

→ In general Plain area

- **Lateral Erosion -** The **widening of a stream** channel by eroding its banks.
- **Headward Erosion -** Erosion at the source of a stream, **extending the stream channel upstream.**
  → Top of head ↳ sources can expand. Eg. Ganga
- **Hydraulic Action (Pressure) -** The **mechanical loosening and removal of material** by the force of river water.
- **Braiding -** Braiding occurs when the **main river channel splits into multiple, narrower channels.** This results in a **network of river channels separated by small, often temporary, islands** called **braid bars.** Braided rivers are common in areas with a low slope and/or a high sediment load. ↳ Majuli Island, Brahmaputra

Erosional Landforms by Running Water:

Valleys:

Valleys represent a major erosional landform, evolving through various stages over time. The initial stage of valley formation begins with **small and narrow rills,** which are **tiny channels** carved into the landscape by flowing water. As these rills are subjected to **continuous erosion, they gradually expand into larger and wider channels** known as **gullies.** With ongoing erosion, **gullies further deepen, widen,** and **lengthen,** eventually transforming into **valleys.**

Types of Valleys:

- **V-Shaped Valleys -** As the name suggests, these valleys have a **pronounced V-shape** when viewed in cross-section. They are typically formed in the **upper course of a river** where the flow of water has a high velocity, cutting sharply into the landscape.

V → Shaped of Erosion

- **Gorge -** A gorge is a type of valley that is **extremely deep with very steep to nearly vertical sides.** Unlike other valleys, a gorge maintains a relatively **consistent width from its top to its bottom.** Gorges typically form in **areas where the bedrock is very hard and resistant to erosion.**

- **Canyon -** A canyon is similar to a gorge but has distinct characteristics. It features **steep, step-like side slopes** and is generally as deep as a gorge. However, a canyon is **wider at its top than at its bottom.** Essentially, a canyon is a variant of a gorge but is commonly formed in **areas with horizontally bedded sedimentary rocks.** Eg. Grand Canyon

Eg. Gandikota canyon → Pennar River → Eastern Ghat

Potholes and Plunge Pools:

- **Potholes -** They are **circular depressions found in riverbeds,** created by the erosive action of flowing water combined with the **abrasion caused by rock fragments.** When a small depression forms, pebbles and boulders get trapped and rotated by the water, gradually enlarging the depression.

- **Plunge Pools -**They are **large potholes that are significantly deeper and wider.** They form at the base of waterfalls due to the intense impact of falling water and the swirling motion of boulders. Plunge pools play a crucial role in the **deepening of valleys by continually eroding the bedrock.**

Waterfalls and Rapids:

- **Waterfalls -** It occur where a **river makes a steep descent over rocky slopes or ledges.** As the river plunges down with great velocity, it **causes significant erosion,** particularly at the base of the waterfall, where **plunge pools** are formed. These waterfalls often occur where the **river flows from softer rock, such as limestone or sandstone, to harder rock like granite.** The softer rock erodes more easily, creating a steep drop over the more resistant rock. Waterfalls, also known as cascades, are spectacular features found in various locations around the world. **Example - Niagara Falls** at the USA-Canada border, **Jog Falls** in India, and **Victoria Falls** in Zambia.

- **Rapids -** It form in areas of shallow, **fast-flowing water,** typically in **younger streams.** They are characterized by turbulent water and a series of small waterfalls or cascades within the stream. These features are common in river channels where the **water flows over a rocky bed, creating a series of drops and waves.** Rapids are **popular sites for adventure sports** like white-water rafting, with notable locations such as Rishikesh in India offering thrilling experiences for enthusiasts.

→ snake like structure

These are due to alternate soft & hard rock bed. Soft rock erode fast than hard rock.

Incised or Entrenched Meanders:

↳ Generally occurs in flood plain where lateral erosion

Special type Meander is Incised Meander

- A **meander** is defined as a **pronounced curve or loop** in the course of a river channel. **Incised or entrenched meanders** are deeply embedded meanders **found in hard rock formations.** They occur where rivers experience **rapid vertical erosion, cutting deeply into the riverbed without significantly eroding the lateral sides.**

- As the river continues to cut into the bedrock, these loops become deeper and wider, eventually forming deep **gorges** or **canyons.** Meandering courses are typically found over floodplains and delta plains where the stream gradients are very gentle, promoting lateral erosion and creating wide, looping bends.

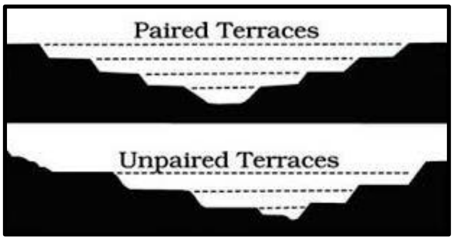
- **Oxbow Lake -** Sometimes, because of intensive erosion action, the **outer curve of a meander** gets accentuated to such an extent that the inner ends of the loop come close enough to get **disconnected from the main channel and exist as independent water bodies** called as **'oxbow lakes.'** These water bodies are converted into swamps in due course of time. In the Indo-Gangetic plains, southwards shifting of Ganga has left many oxbow lakes to the north of the present course of the Ganga.

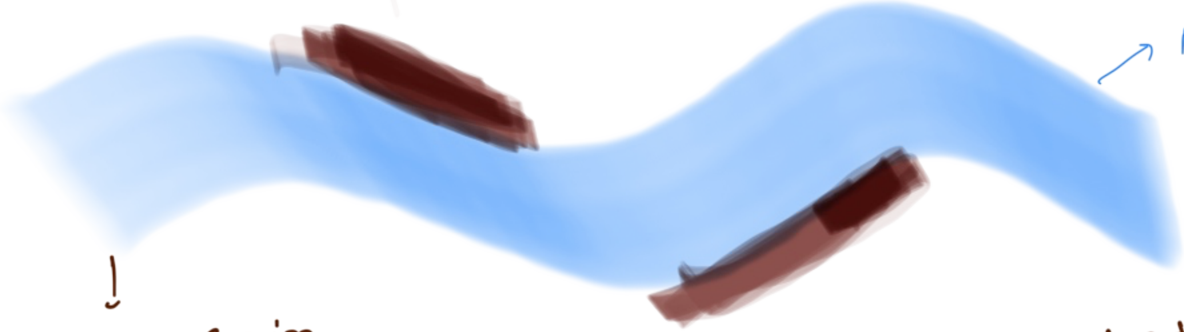
eg. Kanwar lake → river Gandak

River Terraces:

- River terraces are **flat surfaces** that mark former levels of a **river's valley floor or floodplain.** They are primarily erosional features, **created by the vertical erosion of a river cutting down into its own previous floodplain deposits.** River terraces can be found at various elevations along a river valley.

- **When terraces are at the same elevation on both sides** of the river, they are known as **paired terraces.** In contrast, **unpaired terraces occur when terraces are present on only one side** of the river or at different elevations on opposite sides.

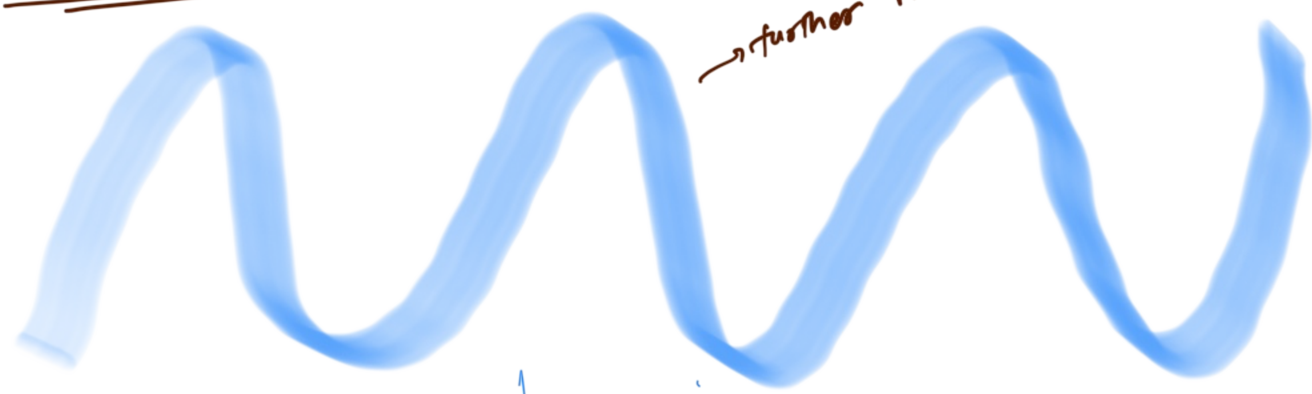




Manders & sedimed deposition

outbank Erosion
other side deposition

further Meandar
due to deposition at one side than other



overtime

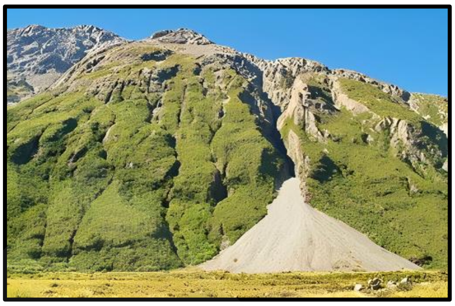
Sediment deposit

This part seperate

This is oxbow lake

Depositional Landforms by Running Water:

Alluvial Fans and Cones:



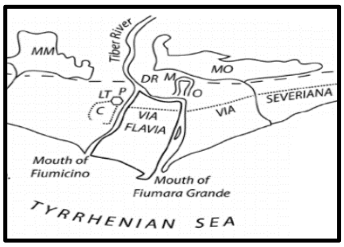
- Alluvial fans are distinctive, **fan-shaped deposits of sediment** that form where a high-gradient stream loses energy as it flows onto a flatter plain, typically at the **base of mountain slopes.** These formations occur when **streams carrying coarse sediment descend steep mountain slopes and suddenly encounter the lower gradient of a foot slope plain.** The abrupt decrease in slope reduces the stream's carrying capacity, causing it to deposit the heavier sediment it can no longer transport. This process results in a broad, **cone-shaped accumulation of coarse materials,** creating what is known as an **alluvial fan.**
- In **humid regions, alluvial fans usually have low cones with gentle slopes,** while in **arid and semi-arid climates,** they appear as **high cones with steep slopes.** The streams on these fans frequently shift their positions, forming multiple channels called **distributaries.**

Deltas: → Generally triangle based shape

- Deltas are **sedimentary deposits** that form at the **mouths of rivers where they flow into a standing body of water, such as an ocean, sea, or lake.** Similar to alluvial fans in terms of their depositional nature, deltas develop in a different environmental setting. As rivers carry sediment towards their mouths, the **reduction in flow velocity causes sediment to settle and accumulate, forming a delta.** Unlike alluvial fans, delta deposits are typically **well-sorted and exhibit clear stratification** due to the continuous action of waves, tides, and currents that redistribute the sediments. Deltas often have a **variety of sub-environments,** including **distributary channels, levees,** and **floodplains,** each contributing to the complexity of the deltaic system.
- Types of Deltas:
  - **Arcuate Delta (Fan-shaped Delta) -** Arcuate deltas are **triangular or fan-shaped** with a convex seaward margin. They are formed in **regions with a high sediment load and low wave** and tidal energy, which allows the river to spread out and deposit sediments evenly.
    - A prime example of an arcuate delta is the **Nile Delta in Egypt.**

  - **Bird's Foot Delta -** Bird's foot deltas resemble the **claws of a bird's foot,** with several distributaries **extending out into the sea.** These deltas develop where **river-dominated processes prevail over wave and tidal actions,** allowing distributaries to extend far out into the water body.
    - **The Mississippi River Delta in the USA** is a notable example.

  - **Cuspate Delta -** Cuspate deltas have a **tooth-like or cusp-shaped structure** with a pointed seaward margin. They form where **strong wave action redistributes sediments** on either side of the river mouth, shaping the delta into a pointed structure.
    - The **Tiber River Delta** in Italy is an example of a cuspate delta.



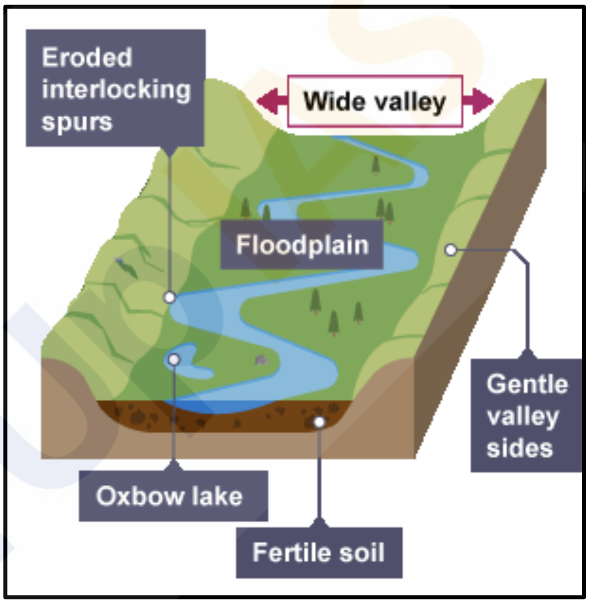
  - **Estuarine Delta -** Estuarine deltas form **within the estuary of a river.** They develop where a river deposits sediment into a long, narrow estuary, and tidal processes significantly shape the delta.
    - **The Seine River Delta** in France is a typical example of an estuarine delta.

↳ Normade River in Arabian sea form Aliabet Estuary

- **Conditions Favourable for Delta Formation**:
  - **Active Erosion -** Extensive vertical and lateral erosion in the upper course of the river provides sediments for delta formation.
  - **Sheltered Coast -** The coast should be sheltered and preferably tideless to allow sediment accumulation.
  - **Shallow Sea -** The adjoining sea should be shallow to prevent sediment dispersion in deep waters.
  - **No Large Lakes -** The river should not have large lakes that filter out sediments.
  - **No Strong Currents -** Strong currents running at right angles to the river mouth should be absent to avoid washing away sediments.

Floodplains:

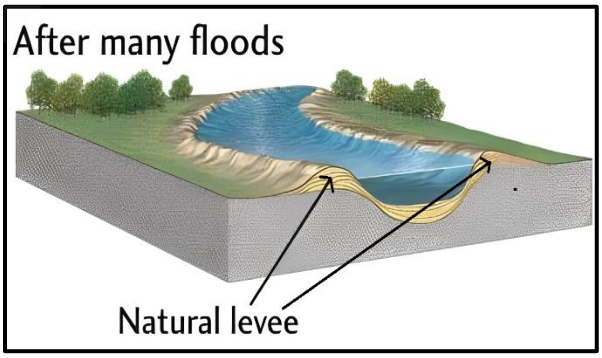
- Floodplains are significant landforms created by river deposition. When a **stream flows from a steeper to a gentler slope, it loses energy and deposits large-sized materials first.** As the water slows down further, it carries finer materials like sand, silt, and clay, **depositing them over the riverbed and spilling over the banks during floods.**



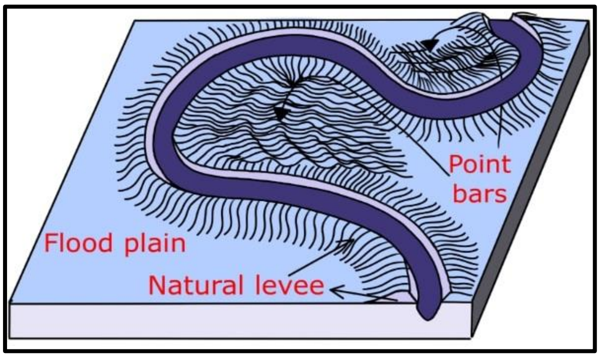
- **The floodplain above the bank,** known as the **inactive floodplain,** contains **two main types of deposits –**
  - **Flood deposits**
  - **Channel deposits.**
- In these areas, **river channels often shift laterally,** changing their courses and leaving behind cut-off channels. These abandoned channels gradually fill with coarse deposits, contributing to the floodplain's structure.
- Floodwaters typically **carry finer materials, such as silt and clay,** which are deposited across the floodplain. When these floodplains are part of a delta, they are specifically referred to as **delta plains.**

Natural Levees and Point Bars:

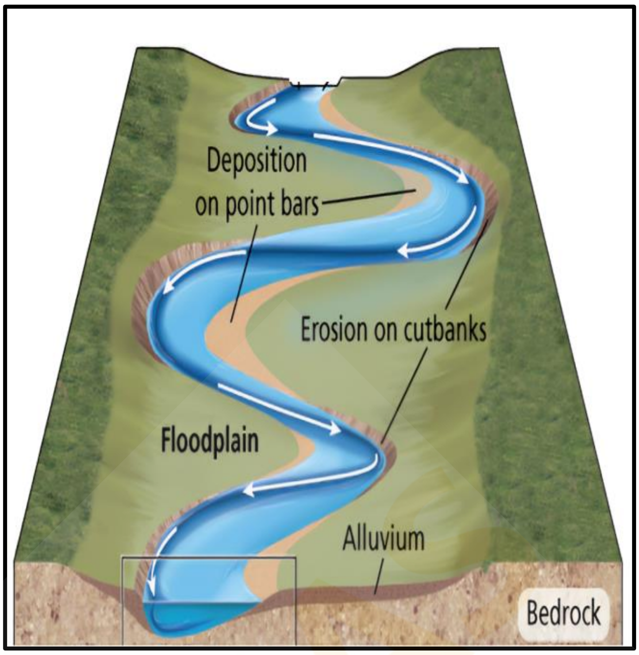
- **Natural levees** - These are the prominent features **found along the banks of large rivers.** These landforms are characterized by **low, linear ridges of coarse deposits that run parallel to the riverbanks.** During periods of flooding, rivers overflow their banks, and the **heaviest sediments are deposited closest to the river, gradually building up these ridges.** Over time, the levees may be cut into individual mounds due to erosional processes, but they typically remain as continuous barriers that help contain the river's flow.



- **Point bars -** Also known as **meander bars,** are formed on the **inner, concave side of river meanders.** As the river flows around bends, it erodes the outer banks and deposits sediments on the inner banks. These deposits **accumulate in a linear fashion, creating point bars that are almost uniform in profile and width.** The sediments within point bars vary in size, including a mixture of fine and coarse materials. **Point bars are integral to the meandering process** of rivers, contributing to the dynamic reshaping of the river channel and floodplain over time.



Meanders:



- In large floodplains and delta plains, rivers often exhibit **loop-like patterns known as meanders.** Meander is **not a landform** but is only a **type of channel pattern.**
- When water flows over areas with very gentle slopes, it tends to work laterally rather than vertically. This **lateral movement exerts pressure on the banks of the river, causing erosion.** The force of the water pushes against the banks, carving out curves and bends, leading to the formation of **meanders.**
- The banks of rivers with meandering channels are often composed of **unconsolidated alluvial deposits.** These materials are loose and easily eroded, making them susceptible to the lateral pressure exerted by flowing water. Any irregularities or weaknesses in these deposits are exploited by the river, contributing to the development of meanders.
- **The Coriolis force,** which affects the movement of fluids on a rotating Earth, also **plays a role in the formation of meanders.** This force causes the water to be deflected as it flows, similar to how it influences wind patterns. The deflection enhances the lateral movement of the water, promoting the meandering pattern.
- The tendency for a river to meander is influenced by the **balance between erosion and deposition.** If there is no significant deposition or erosion, the river is less likely to develop pronounced meanders. **Erosion on the outer banks of bends and deposition on the inner banks** (creating point bars) are crucial processes that sustain the meandering pattern.

Landforms Made by Groundwater:

Groundwater acts as a significant erosional force, capable of dissolving solid rock over time. It performs **erosion, transportation,** and **deposition,** leading to the creation of several picturesque topographical features. This process begins when **rainwater absorbs carbon dioxide (CO_2)** as it falls through the atmosphere. The absorbed CO_2 combines with the water to form a **weak carbonic acid solution.** When this slightly acidic water infiltrates the ground**, it percolates through the pore spaces in the soil and the cracks and fractures in the underlying rock.** This subsurface water movement is what we refer to as groundwater.

One of the most notable effects of groundwater is its ability to **dissolve limestone.** Limestone, composed mainly of **calcium carbonate,** reacts readily with carbonic acid. As groundwater flows through limestone formations, it gradually dissolves the rock, creating various landforms characteristic of regions known as **'Karst topography.'**

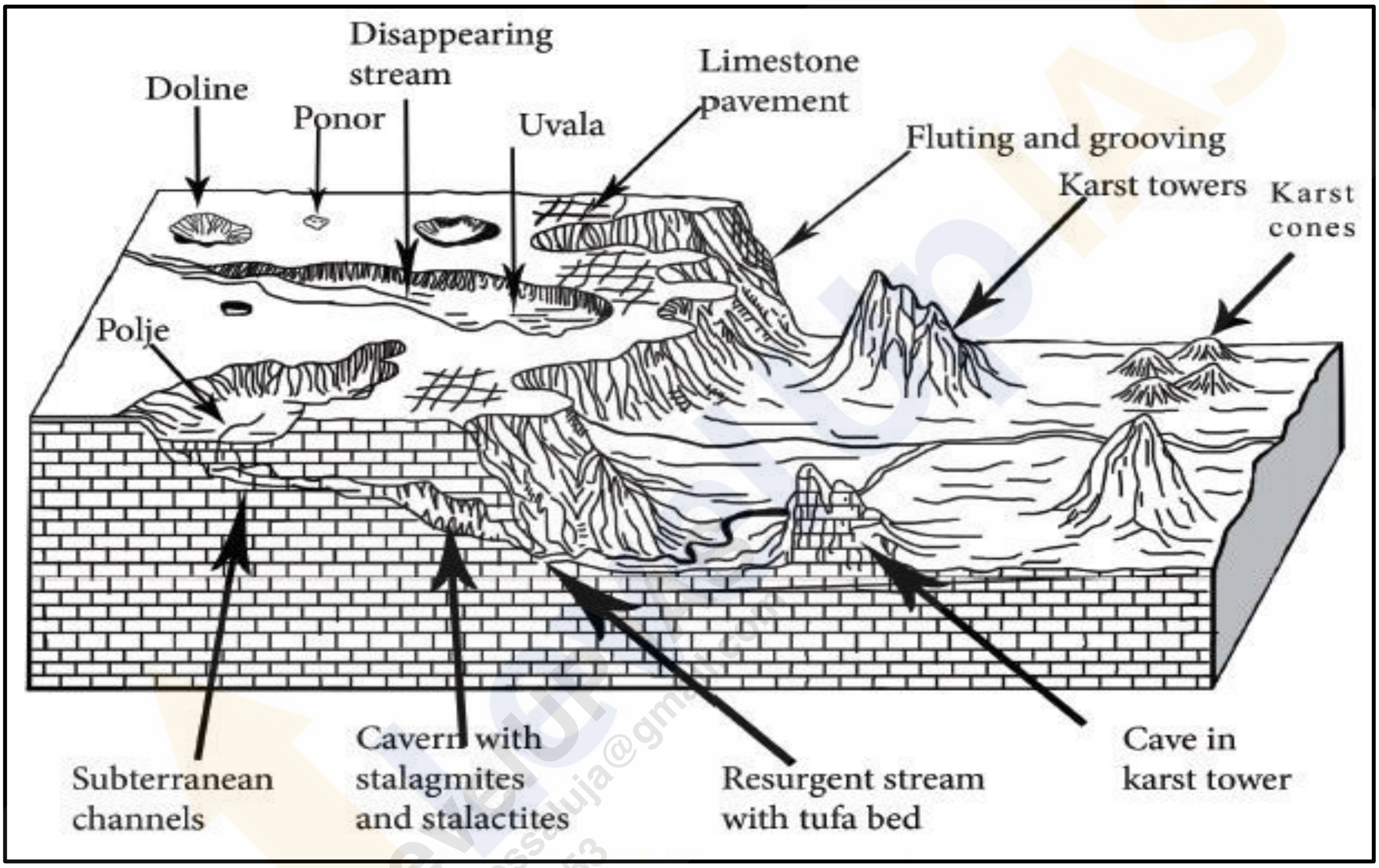
Karst Topography:

The term **"Karst"** is derived from the **Karst region** along the **Adriatic Sea coast in Croatia** (formerly Yugoslavia), where such formations are prominent. Karst topography is identified in areas where the landscape is shaped predominantly by the **dissolution of soluble rocks, primarily limestone and dolomite.**

Karst landscapes are found in various regions around the world, including the **Causses of France, the Kwangsi area of China, the Yucatán Peninsula in Mexico,** and several locations in the **United States** such as the **Midwest, Kentucky,** and **Florida.**

In India, karst topography is present in the **Vindhya region** (particularly southwestern Bihar), **parts of the Himalayas** (including Jammu & Kashmir, Robert Cave, Sahasradhara, the eastern Himalayas, and areas near Dehradun), **Pachmarhi** in Madhya Pradesh, **Gupt Godavari Cave** in Chitrakoot (Uttar Pradesh), the coastal areas near **Vishakhapatnam** (**Borra Caves**), and **Bastar** in Chhattisgarh and **Borra Caves** in Andhra Pradesh.

The action of groundwater through the processes of **solution** (dissolution of rock) and **deposition** (precipitation of minerals from groundwater) leads to the formation of unique landforms in these regions. These landforms can be broadly classified into **erosional and depositional types.**



Erosional Landforms by Groundwater:

Swallow Holes:

- Small to medium-sized **round to sub-rounded shallow depressions,** known as swallow holes, form on the surface of limestone through the process of solution. These features are common in limestone and karst areas.

Sinkholes (Doline):

- Sinkholes are prevalent in limestone or karst regions. They are openings that are more or less **circular at the top and funnel-shaped towards the bottom,** with sizes ranging from a few square meters to a hectare, and depths varying from less than half a meter to over thirty meters.

- Sinkholes can form solely through **solution action** (solution sinks) or may start as solution features. If the bottom of a sinkhole forms the roof of an underground void or cave, it might collapse, leaving a large hole opening into the cave or void below, known as **collapse sinks.** Solution sinks are more common than collapse sinks.
- Quite often, sinkholes are covered with a soil mantle and appear as shallow water pools. Anyone stepping over such pools might sink, similar to quicksand in deserts. The term **"doline"** is sometimes used to refer to collapse sinks.

Valley Sinks or Uvalas:

- When **sinkholes and dolines join together due to slumping of materials** along their margins or **roof collapse of caves, long, narrow to wide trenches** called **valley sinks or uvalas form.** Gradually, most of the limestone surface is eaten away by these pits and trenches, creating an extremely irregular surface with a maze of points, grooves, and ridges known as lapies.

Lapies and Limestone Pavements:

- Lapies are **ridges formed due to differential solution activity along parallel to sub-parallel joints.** These features can appear as grooves, flutes, or ridge-like formations in an open limestone field. Over time, these lapie fields may eventually transform into somewhat smooth limestone pavements.

Caves or Caverns:

- In areas with **alternating beds of rocks** (such as shales, sandstones, and quartzites) with limestones or dolomites in between, or in regions where limestones are dense, massive, and occur as thick beds, cave formation is prominent. **Water percolates down through the materials or through cracks and joints, moving horizontally along bedding planes.** It is along these bedding planes that limestone dissolves, resulting in long and narrow to wide gaps called **'Caves.'**
- Caves may form at different elevations depending on the limestone beds and intervening rocks, creating a maze of caves. Typically, caves have an **opening through which cave streams are discharged.** Caves with openings at both ends are referred to as **tunnels.**

Polje:

- A polje, also called a karst polje or karst field, is a **large, flat-bottomed depressions formed by the collapse of a large underground cavern or by the dissolution of surrounding rock.** These features are distinguished by their vertical side walls, **flat alluvial floors, independent drainage systems within the basin, irregular borders, and often a central lake.** Poljes are essentially large, closed basins with an elliptical shape, which can cover areas up to 258 square kilometers. They are commonly found in the karst regions of the former Yugoslavia and Jamaica.

Depositional Landforms by Groundwater:

Limestone caves are known for their fascinating depositional landforms, primarily formed from **calcium carbonate.** The key processes involved include the **dissolution** and **deposition** of calcium carbonate from carbonated water, which contains dissolved carbon dioxide.

Stalactites: ceil

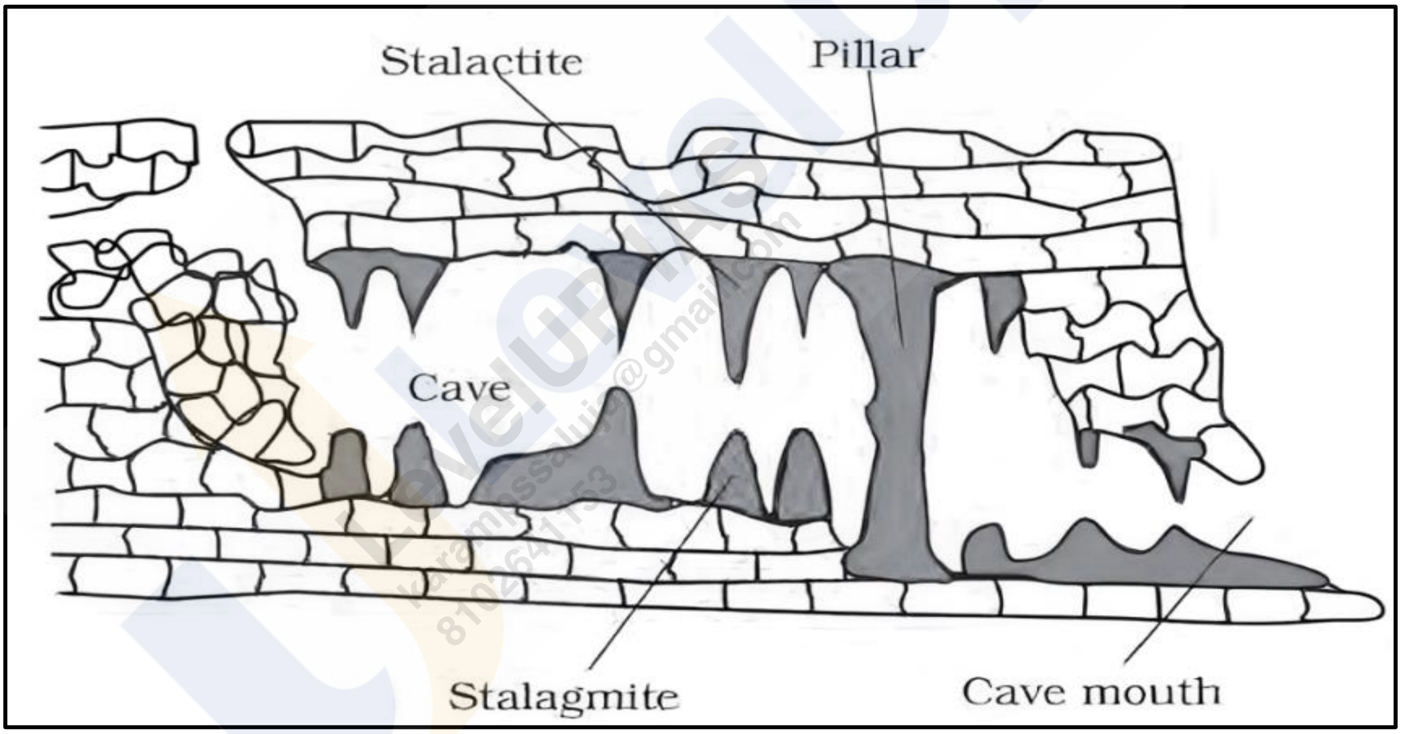
- Stalactites are **icicle-shaped** formations that **hang from the ceilings of caves.** They vary in diameter, generally being wider at their bases and tapering towards the ends. Stalactites form when **water dripping from the cave's ceiling carries dissolved calcium carbonate.** As the **water evaporates or loses its carbon dioxide, calcium carbonate is deposited,** creating these hanging formations.

Stalagmites: ground

- Stalagmites **rise from the cave floor directly beneath stalactites.** They develop from the dripping water that falls from the ceiling or from the thin pipe of the stalactite above. Stalagmites can take various shapes, including columns, discs, or rounded forms with smooth, bulging ends. They may also have a crater-like depression at the top.

Pillars:

- Pillars are formed when **stalactites and stalagmites grow together and merge, creating large, vertical columns.** These formations can vary widely in diameter and shape, reflecting the continuous deposition of minerals from both the ceiling and the floor.



Land form by G. Water

E
- Sinkhole
- Swallowhole
- Dolin
- Uvala
- cave

D
- Pillar
- Stalactite
- stalaqmite

KARST TOPOGR

- Water table below limestone
- Rock Joint
- Arid semi-arid region
- Huge volume of thick bedded limestone

Lf Glacier

Erosion
(1) Tarn, Cirque OR corre
(2) Ushaped valley
(3) Hanging
(4) Pyramidical peak
(5) Arete

D
- Eske
- Moraine
- Drumlin {egg shaped}
- outwash plain
- Till plain

Landforms Made by Glaciers:

Glaciers are **massive, moving bodies of ice,** often referred to as **"rivers of ice."** They form on land where climatic conditions support the accumulation of snow and ice. This occurs in regions where the seasonal snowfall exceeds the seasonal melting. Over time, **layers of snow accumulate, compact, and recrystallize into ice,** leading to the formation of glaciers.

Types of Glaciers:

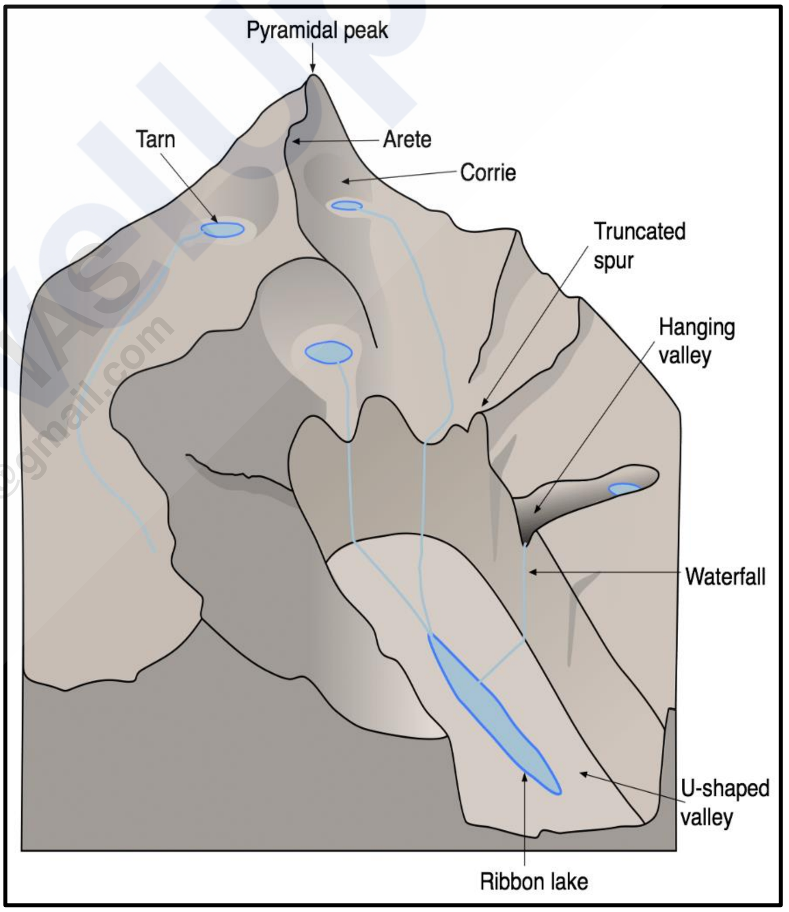
Continental Glaciers:

- Continental glaciers, also known as **ice sheets, are thick ice masses that cover vast areas of land.** They form in **polar regions** and can reach **thicknesses of several thousand meters.** Continental glaciers primarily build up at their centers and spread outward in all directions. **Examples - Greenland Ice Sheet, Antarctic Ice Sheet** and **Arctic Ice Sheets.**

Mountain or Valley Glaciers:

- Mountain or valley glaciers form in **high mountain regions and flow down existing valleys.** Their size and shape are influenced by the valley's slope and width. **Examples - Fedchenko Glacier, Siachen Glacier** and **Gangotri Glacier.**

Erosional Landforms by Glaciers:



Cirques or Corrie:

- Cirques are common landforms in glaciated mountains, often **found at the heads of glacial valleys.** They are **deep, long, and wide troughs or basins with steep, concave walls.** Cirques are formed by the erosion of ice as it moves down the mountainside. After the glacier melts, a **cirque lake, or tarn,** may form within the cirque. Sometimes, multiple cirques can be arranged in a stepped sequence.

Glacial Valleys or U-Shaped Valleys:

- Glacial valleys are **U-shaped troughs** with **broad floors and steep, smooth sides.** These valleys may contain debris or moraines and can have lakes formed by glacial meltwater. In some cases, **these glacial troughs can be inundated with sea water, forming fjords - deep, narrow sea inlets** typically found in high-latitude regions. Fjords have strikingly steep cliffs and are a dramatic example of glacial erosion.

Hanging Valleys:

- Hanging valleys are **formed by tributary glaciers that join larger glaciers.** These tributaries have less ice and, therefore, less erosive power, resulting in **valleys that are higher than the main glacier valley.** Waterfalls often form at the junction of hanging valleys and main valleys.

Horns and Serrated Ridges (Aretes):

- **Horns -** These are **sharp, pointed peaks** formed where **three or more cirques erode the mountain headward** until they converge. The intersecting walls create a steep, pointed peak.
- **Serrated Ridges (Aretes) -** These are **narrow, saw-toothed ridges** formed between cirques due to **progressive erosion.** They have a sharp crest and a zig-zag outline.

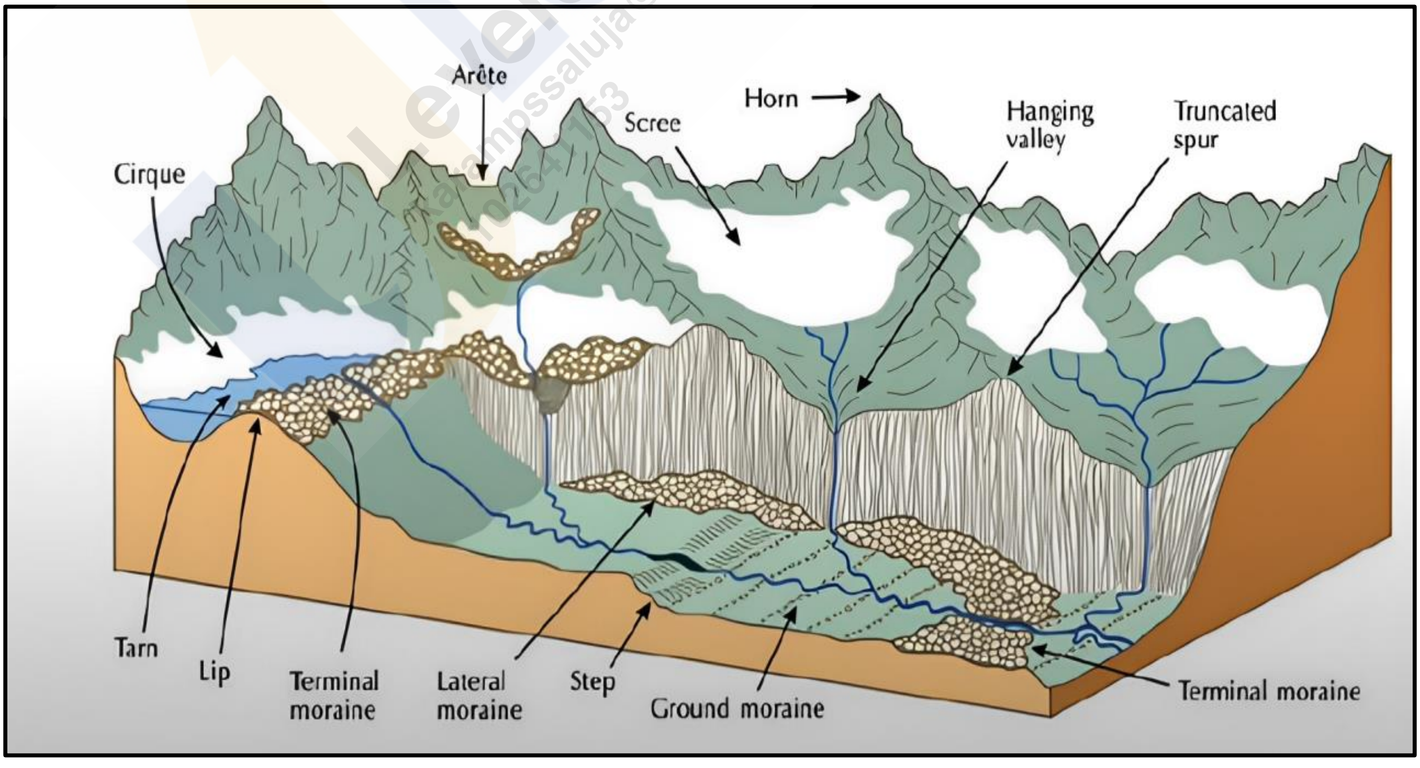
Depositional Landforms by Glaciers:

Glacial Till:

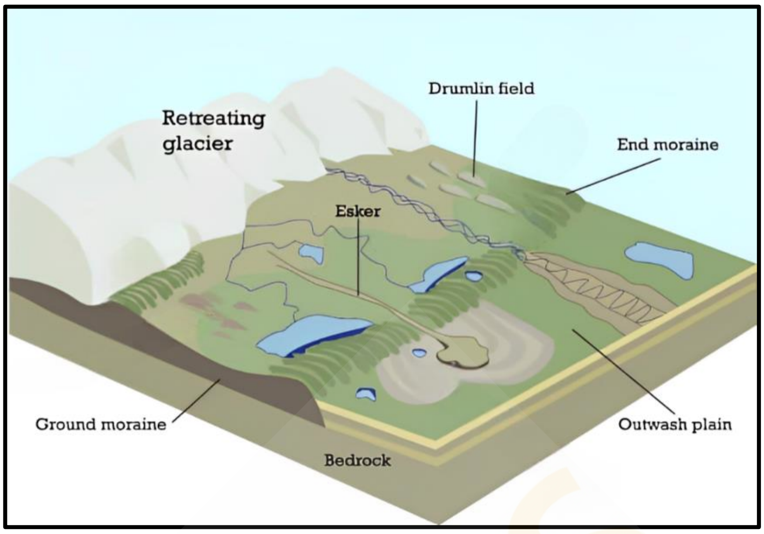
- Glacial till refers to the **unsorted and unstratified material deposited directly by melting glaciers.** This material includes a **mix of coarse and fine debris.** Some finer debris is transported by meltwater streams, which carry and deposit it further downstream. These deposits are known as outwash deposits, which are characterized by a more stratified and sorted structure compared to glacial till.

Moraines:

- Moraines are **elongated ridges formed from the accumulation of glacial till.** They are categorized based on their position relative to the glacier:
  - **Terminal Moraines -** These are ridges of **debris deposited at the furthest advance of the glacier,** marking its maximum extent.
  - **Lateral Moraines -** These ridges **run parallel to the sides of a glacier** and are formed from material accumulated along the glacier's edges.
  - **Ground Moraines -** When a **glacier retreats, it often leaves behind a sheet of till covering the valley floor,** known as a ground moraine. This deposit can be **irregular and uneven.**
  - **Medial Moraines -** Located in the **center of a glacial valley between two lateral moraines,** medial moraines are created from the merging of two lateral moraines. They may sometimes appear similar to ground moraines due to their less distinct formation.



Eskers:



- Eskers are **long, winding ridges composed of sand and gravel** that form from sediments **deposited by meltwater** flowing through tunnels or channels beneath or on top of a glacier. As the glacier melts, the sediment left behind creates these **long, narrow ridges.** These ridges often follow the direction of glacial flow and can extend for over 100 kilometers.

Drumlins:

- Drumlins are **elongated, oval-shaped hills formed from glacial drift beneath a glacier or ice sheet.** They are aligned in the direction of the glacier's flow and are typically low hills up to 1.5 kilometers in length and 60 meters in height. Drumlins are **steeper on the side facing the glacier's advance and taper off on the opposite side.** When found in clusters, this feature is known as '**basket of eggs' topography.**

Outwash Plains:

- Outwash plains are **flat areas composed of sediment washed out from terminal moraines** by meltwater streams. The meltwater sorts and redeposits materials such as sand and fine sediments, creating plains that are often **suitable for specialized agriculture** due to their sandy soils.

Kettle Lakes:

- Kettle lakes are depressions **formed in outwash plains when blocks of ice become buried in sediment and later melt.** These shallow, sediment-filled bodies of water are remnants of the glacial ice that once covered the area.

Landforms Made by Waves and Currents or Coastal Landforms:

Coastal processes are among the **most dynamic and often the most destructive forces** shaping our planet. Understanding these processes is crucial, as they can bring rapid and dramatic changes to coastal landscapes. Coastal changes can be both erosional and depositional, often varying with the seasons.

Most coastal changes are driven by the action of waves. When **waves break, they hurl water with tremendous force against the shore, causing a churning of sediments on the sea bottom.** This constant impact significantly alters coastlines. Storm waves and tsunamis can affect even more drastic changes in a shorter period compared to normal waves. As the wave environment fluctuates, so does the intensity of breaking waves.

Factors Influencing Coastal Landforms:

- **Configuration of Land and Sea Floor -** The physical **layout of the coast and the underwater topography** play a crucial role.
- **Coastal Movement -** Whether the coast is **advancing** (emerging) seaward or **retreating** (submerging) landward.

Assuming sea level remains constant, two primary types of coasts help explain the evolution of coastal landforms:

1. **High, rocky coasts** (submerged coasts).
2. **Low, smooth, and gently sloping sedimentary coasts** (emerged coasts).

High Rocky Coasts:

- High rocky coasts are characterized by **drowned river valleys and highly irregular shorelines, often indented with fjords.** These coastlines typically show **steep hillsides** that plunge sharply into the water, with **erosion features being dominant initially** and minimal depositional landforms present.
- Along high rocky coasts, **waves crash with great force, sculpting the hillsides into cliffs.** Over time, the relentless pounding of waves causes these cliffs to recede, leaving behind a **wave-cut platform** in front of the sea cliff. This process gradually smooths out the shoreline's irregularities.
- The materials eroded from the sea cliffs break into **smaller fragments, become rounded, and are deposited offshore.** With ongoing cliff erosion and material deposition, a wave-built terrace eventually forms in front of the wave-cut terrace. The eroded materials also provide a supply for longshore currents and waves, which deposit them as beaches along the **shore and as bars** (long ridges of sand and/or shingle parallel to the coast) in the nearshore zone. When these bars rise above water, they are **called barrier bars.** A barrier bar attached to the headland of a bay is known as a **spit.** When spits or barrier bars block the mouth of a bay, they create a **lagoon.** Over time, these lagoons can fill with sediments from the land, forming **coastal plains.**

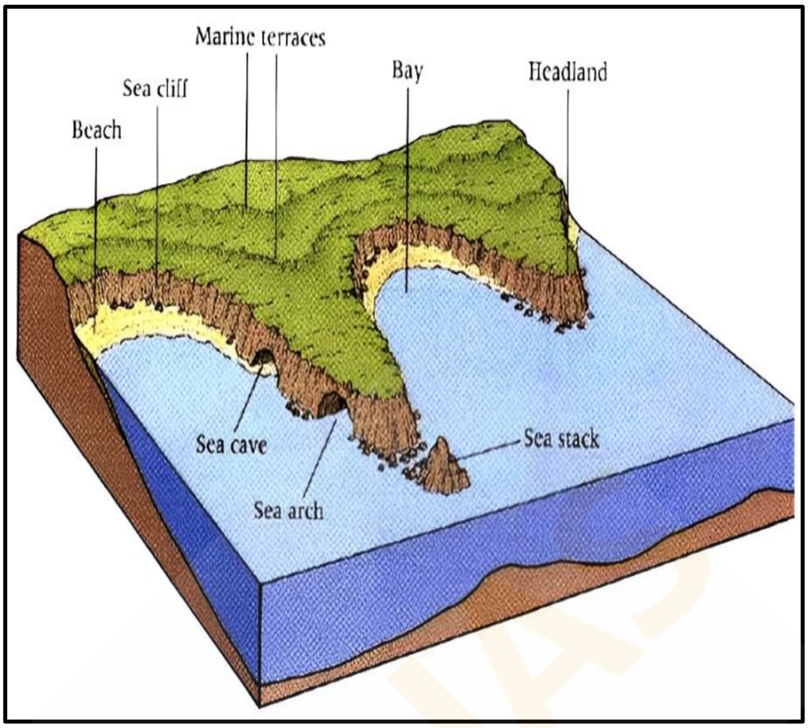
Low Sedimentary Coasts:

- Low sedimentary coasts typically feature **rivers extending their lengths by building coastal plains and deltas.** These coastlines appear smooth, with **occasional incursions of water in the form of lagoons and tidal creeks.** The land slopes gently into the water, and marshes and swamps are common.
- On gently sloping sedimentary coasts, breaking waves churn the bottom sediments, moving them to **build bars, barrier bars, spits, and lagoons.** Over time, lagoons can transform into swamps and eventually into coastal plains. The stability of these depositional features relies on a steady supply of sediments. However, storm waves and tsunamis can cause significant changes regardless of sediment supply. Large rivers that deliver substantial amounts of sediments form deltas along these low sedimentary coasts.

Do you know?

*The **west coast** of our country is **a high rocky retreating coast.** Erosional forms dominate in the west coast. The **east coast** of India is a **low sedimentary coast.** Depositional forms dominate in the east coast.*

Erosional Coastal Landforms:



Sea Cliffs:

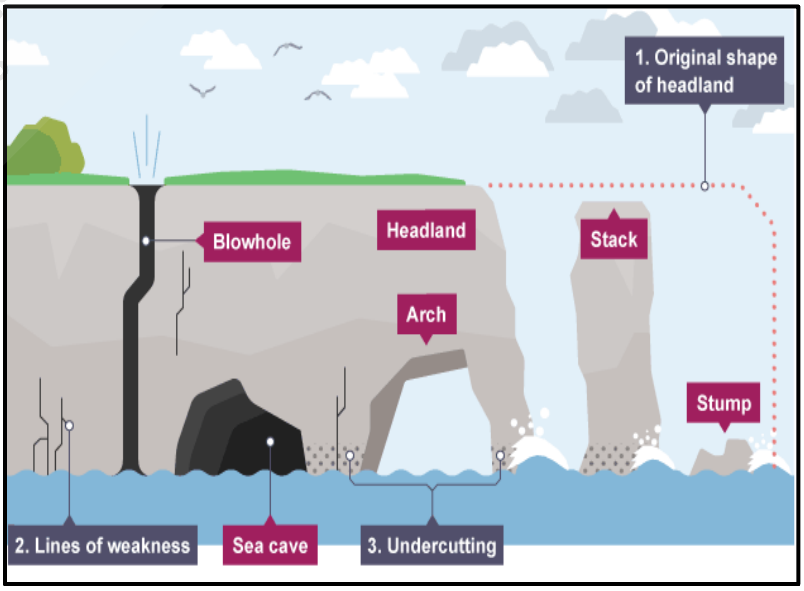
- Sea cliffs are prominent features formed by marine erosion. These **steep, rocky coastlines rise almost vertically above sea level.** The steepness of a cliff is determined by variations in geological structure, lithology, and the relative rates of subaerial weathering and erosion. The **lower part of a cliff experiences the maximum impact from sea waves,** especially if it is composed of softer rocks like limestone, which erode more rapidly than the harder rocks typically found in the upper part. Over time, the **unsupported upper sections project outward and eventually collapse into the sea,** leaving behind a vertical wall known as a **cliff.** Numerous sea cliffs can be observed along the Konkan coast of India.

Sea Caves:

- Sea caves form when waves crash against headlands, **eroding weaker sections such as cracks.** Differential erosion creates a hollow in the **lower part of the rock, especially if it consists of softer material, while the upper part remains harder.** The continuous pounding of waves compresses air inside the hollow, which expands when the waves recede, subjecting the rocks to significant pressure. **Over time, this process enlarges the hollow into a sea cave.** The cave can further expand and deepen, especially if the roof collapses due to the lack of support.

Blowholes:

- Blowholes develop **when sea caves grow larger and waves begin to crash into the back wall of the cave, forcing air and water upwards through the roof.** This process creates a vertical opening, or blowhole, in the headland. Blowholes are rare and require a vertical crack line to be effectively exploited by the erosive forces of waves.



Sea Stacks:

- Sea stacks are **isolated pillars of rock that form when a sea arch collapses.** Initially, sea arches develop from the merging of sea caves that form on opposite sides of a headland. When these arches give way, the remaining rock columns are known as sea stacks.

Sea Terraces:

- Sea terraces are **rock platforms that form when a sea cliff, along with its wave-cut platform, is uplifted above the sea level.** These terraces represent former positions of sea level and are often evident as **flat, bench-like landforms** above the current shoreline.

Depositional Coastal Landforms:

Beaches:

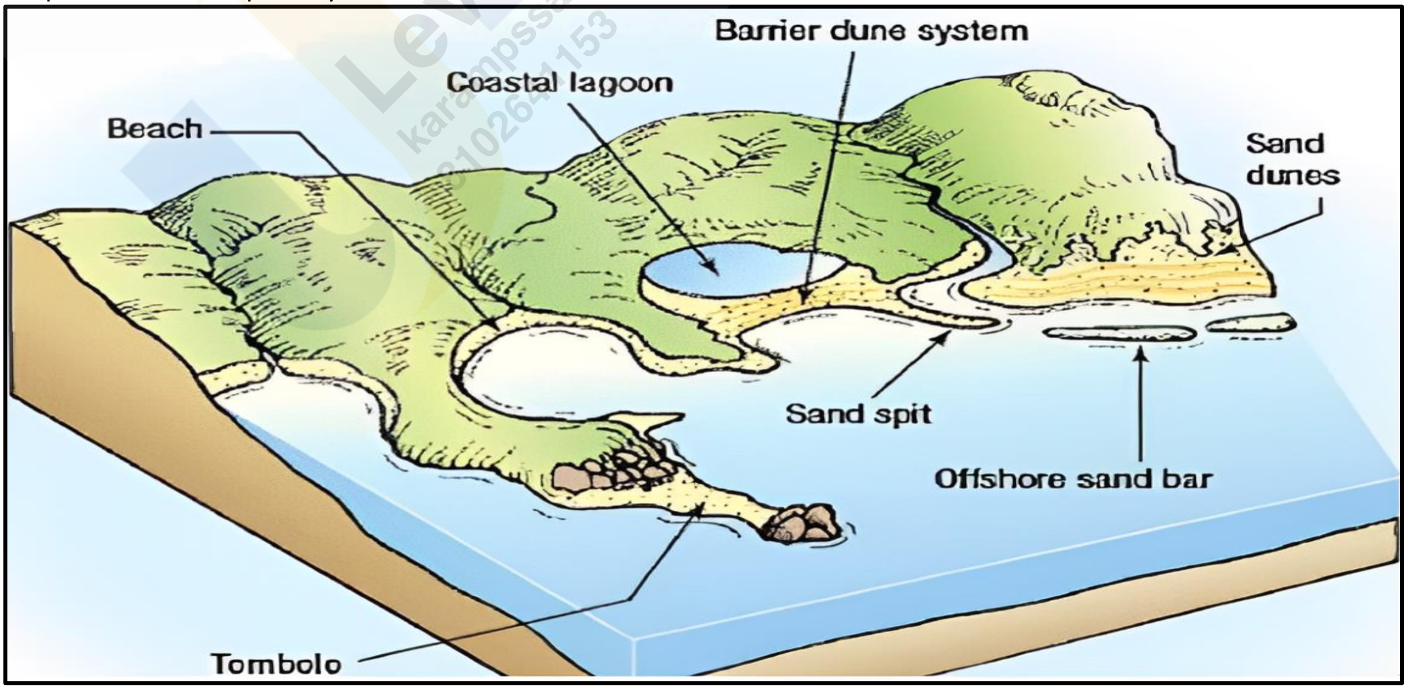
- Beaches serve as **natural barriers** between waves and upland features, such as dunes. They are **narrow, gently sloping patches of land** found along oceans, seas, rivers, and lakes. Composed of **unconsolidated sand, gravel, rocks,** and **seashells,** beaches are constantly reshaped by waves and wind.
- **Calmer weather** and gentle, long-period waves, known as swells, **tend to deposit sediment onshore,** building up the beach. In contrast, **steep waves from storms and strong winds erode the beach by transporting sediment offshore.** Wind blowing across the beach moves finer particles, which accumulate to form dunes when they encounter barriers such as vegetation.
- Notable examples of beaches include Radhanagar Beach in the Andaman and Nicobar Islands, Agonda Beach in Goa, Marina Beach in Chennai, Seminyak Beach in Bali, and Santa Monica Beach in California.

Spits and Bars:

- **Spits -** A sand spit is a **linear deposit of sediment** that extends from the land into the open water. Spits typically form where the **coastline bends inward from the prevailing direction of longshore drift.** The sediment accumulates in the direction of the updrift coast, creating a **narrow, elongated landform.**
- **Bars -** A sandbar, also known as an **offshore bar,** is a **submerged or partially exposed ridge of sand built by wave action offshore from a beach.** These bars are formed by the **deposition of sand** and other sediments, creating a barrier that can be seen during low tide or through the clear waters of shallow coastal areas.

Sand Dunes:

- Sand Dunes are **mounds or ridges of sand deposited by the wind** immediately landward of the beach. **Primary dunes, closest to the beach, are usually lightly vegetated** with grasses, while **secondary dunes farther inland may have denser vegetation,** including shrubs and trees.
- **Vegetation traps and holds windblown sand, stabilizing the dunes.** Large primary dunes provide protection from flooding and wave action for upland areas and can replenish eroded beaches. Secondary dunes offer additional protection if the primary dune is breached.



Tombolos:

- A **sandy isthmus connecting a small island to the mainland.** It is created by the deposition of sediments by waves and tides, which gradually build up to connect the two land masses.

Lagoons:

- Lagoons are **shallow, often brackish water bodies located between barrier bars or spits and the mainland,** formed when sandbars or spits enclose an area of water behind them.

Estuaries:

- Estuaries are **partially enclosed coastal bodies of water,** where freshwater from rivers and streams mixes with seawater. The deposition of sediments from river runoff creates unique habitats.

Mudflats:

- **Soft, muddy sediments** exposed during low tide.

Salt Marshes:

- Coastal wetlands dominated by **salt-tolerant vegetation,** formed by the deposition of fine sediments carried by tidal waters.

Landform Made by Wind or Desert Landforms:

Wind acts as a **geomorphic agent across all terrestrial environments,** with its effects being **most pronounced** in arid regions (Deserts) characterized by fine-textured soils and sediments and sparse or absent vegetation. **A desert is a landscape characterized by extremely low precipitation, leading to harsh conditions for both plant and animal life.** The scarcity of vegetation leaves the ground surface vulnerable to denudation processes. **Approximately one-third of the Earth's land surface is classified as arid or semi-arid,** encompassing regions such as the Polar Regions, which, due to minimal precipitation, are often referred to as polar or **"cold deserts."**

Deserts can be categorized based on various factors including the amount of precipitation, temperature, causes of desertification, or geographical location. For instance, about one-fifth of the world's land is desert. Some deserts are so devoid of life that they are termed **"true deserts."**

Types of Deserts:

- **Hamada (Rocky Desert) -** These deserts are characterized by **extensive areas of bare rock, stripped of sand and dust by wind action.** The exposed rocks are often smooth and polished, resulting in a **highly sterile environment.**

- **Reg (Stony Desert) -** Reg deserts feature large expanses of **angular pebbles and gravels** that are **not easily moved** by wind. These deserts are generally more accessible than sandy deserts and can **support large herds of camels.**

- **Erg (Sandy Desert) -** Also known as **"seas of sand,"** these deserts are dominated by vast stretches of **sand dunes** formed by wind deposition. The dunes often shift with the wind, creating a **dynamic landscape.**

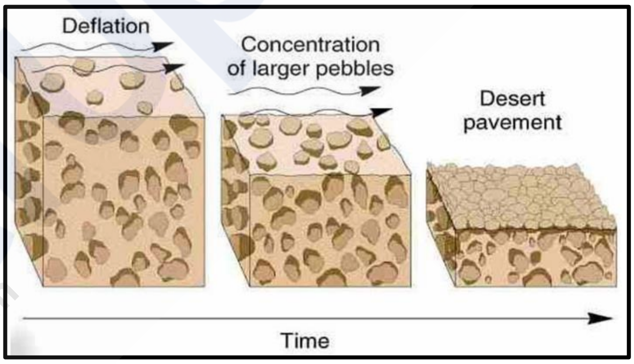
- **Badlands -** Characterized by **gully and ravine formations** on slopes and rock surfaces due to **water erosion,** badlands are **unsuitable for agriculture** and often lead to the abandonment of the region by its inhabitants.

- **Mountain Deserts -** These deserts are found in **highland regions,** such as plateaus and mountain ranges, where erosion has sculpted the landscape into **rugged peaks and uneven terrain.** The steep slopes are often marked by **dry valleys known as wadis,** which have irregular edges shaped by **frost action.**

Action of Wind:

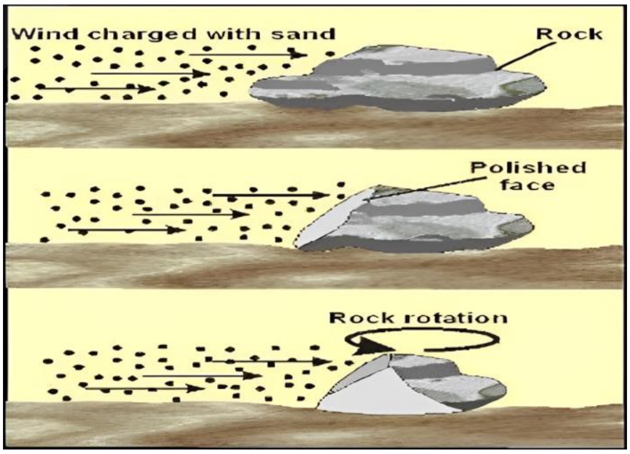
Deflation:

- Deflation involves the **removal of loose surface materials by the wind,** which lifts and carries away these particles. The efficiency of deflation depends on the wind's ability to transport different sizes of material. **Fine dust and sand can be transported over long distances and may even be deposited beyond desert margins.** Over time, deflation leads to the formation of large depressions known as deflation hollows.



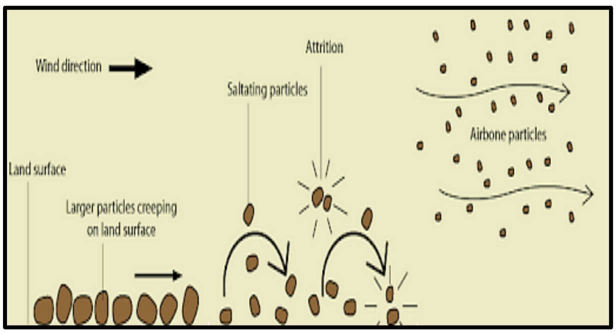
Abrasion:

- Abrasion occurs when **wind-driven sand particles strike rock surfaces, effectively sandblasting them.** This process results in the **scratching, polishing, and wearing away of the rocks.** Abrasion is most intense near the base of rocks, where the wind carries a higher concentration of particles. For this reason, **desert telegraph poles often have metal coverings up to several feet above the ground** to protect them from wind erosion.



Attrition:

- Attrition happens when **wind-borne particles collide with each other during transport.** These collisions cause the **particles to wear down, resulting in smaller, rounded grains** known as **millet seed sand.**



Erosional Landforms by Wind:

Rock Pedestals (Mushroom Rocks):

- Rock pedestals, also known as **mushroom rocks,** are formed by the **sandblasting action of wind against projecting rock masses.** As wind-driven sand **erodes the softer layers of rock,** it leaves behind irregular edges and a distinctive shape. Over time, this process carves grooves and hollows into the rock, resulting in grotesque, pillar-like formations. The **erosion is most intense near the base** where friction is greatest, leading to the development of mushroom-shaped rocks with a broader top and a narrower base.

Mesas and Buttes:

- A mesa, from the Spanish word for **'table,'** is a **flat-topped** landform with **steep sides.** It features a resistant horizontal top layer that protects the underlying rock from erosion by both wind and water. Mesas can be found in canyon regions, such as Arizona, or on fault blocks like Table Mountain in South Africa. Over time, continued erosion can reduce mesas to **isolated flat-topped hills called buttes**, which are characterized by their reduced size and steep sides.

Zeugen:

- Zeugen are **tabular landforms where a hard, resistant rock layer covers a softer underlying layer.** The differential erosion between these layers creates a ridge-and-furrow landscape. Initially, mechanical weathering opens joints in the surface rocks, and wind abrasion further erodes the softer layer, forming deep furrows. The **harder rock stands above these furrows as ridges.** Zeugen can range from 10 to 100 feet in height, and ongoing wind erosion gradually lowers these features while widening the furrows.

Yardangs:

- Yardangs **resemble zeugen but differ in their orientation.** In yardangs, **the hard and soft rock layers are arranged vertically.** Winds, aligned with the direction of the prevailing **winds, erode the softer rock into narrow corridors, leaving behind steep-sided ridges of harder rock.** These long, narrow landforms, called yardangs, are characterized by their distinct, streamlined shapes formed by continuous wind abrasion.

Inselbergs (Island Mountains):

- **Inselbergs, or island mountains,** are isolated **residual hills** that **rise sharply** from the surrounding terrain. Typically composed of **granite or gneiss,** they feature very steep slopes and **rounded tops.** These landforms are remnants of a once extensive plateau that has been largely eroded away.

Pediments and Pediplains:

- **Pediments** are **gently sloping, rocky surfaces** found at the base of **mountains in desert regions.** These landforms develop through the erosion of the mountain front, primarily due to **lateral erosion by streams and sheet flooding.** Initially, erosion begins along the steep margins or sides of tectonically controlled features. As pediments form, they typically start with a steep wash slope and a cliff or free face above. Over time, the wash slope and free face retreat backward, a process known as **parallel retreat of slopes or backwasting**. This retreat extends the pediment backward at the expense of the mountain front, eventually reducing the mountain to a low, featureless plain known as a pediplain.

Playas:

- In **desert basins surrounded by mountains and hills,** the drainage often directs toward the center of the basin, where sediment gradually accumulates to form a nearly level plain. During periods of sufficient moisture, this plain may be covered by a **shallow water body.** These shallow lakes are termed playas and typically **retain water only temporarily due to high evaporation rates.** Playas often **accumulate salts,** and when these salts cover the playa plain, it is referred to as an alkali flat.

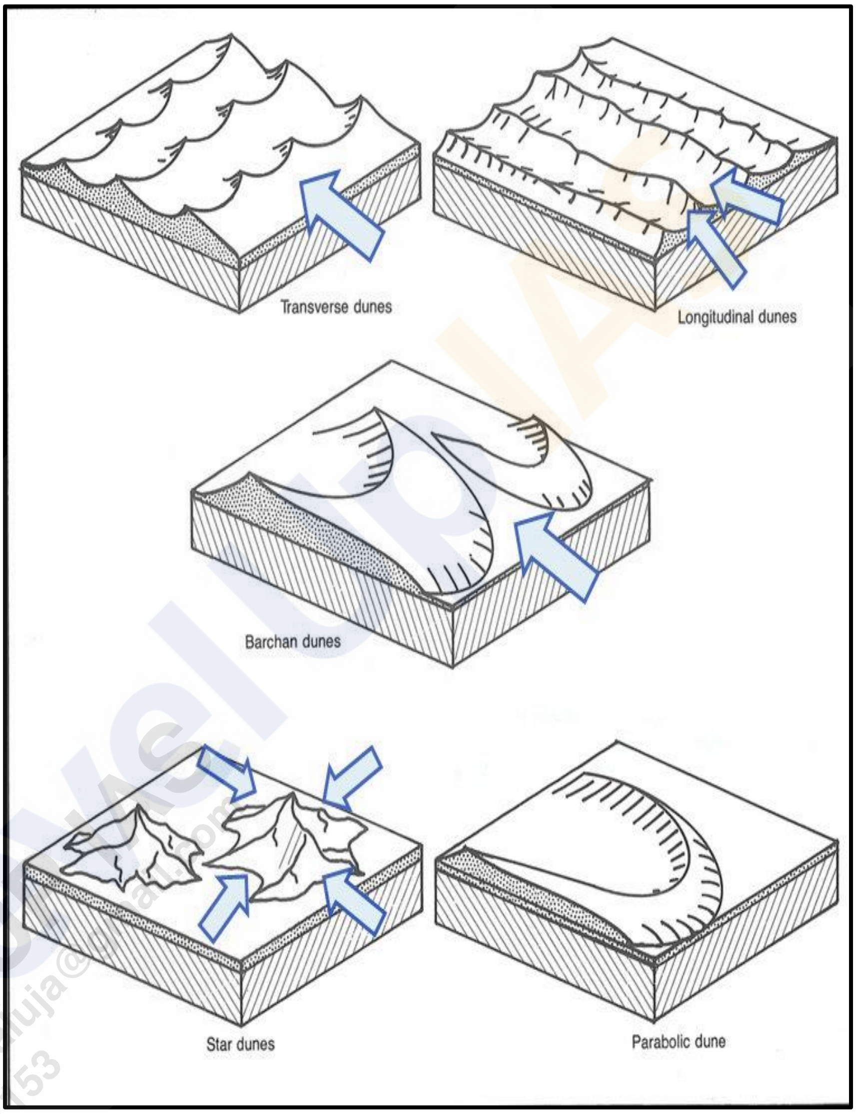
Deflation Hollows and Caves:

- **Deflation Hollows -** Also known as blowouts, these are depressions formed by the removal of loose particles from the ground by wind. Typically small in scale, blowouts can occasionally expand to several kilometers in diameter.
- **Caves -** As wind-driven sand and other particles continually impact rock faces within deflation hollows, these depressions can deepen and widen, evolving into cave-like structures.

Depositional Landforms by Wind:

Dunes:

- Dunes are hills of sand formed by the accumulation and movement of sand shaped by the wind. They can be classified as either active (constantly moving) or fixed (stabilized by vegetation). Dunes are most prominent in erg deserts, where vast seas of sand are continuously reshaped by wind.
- **Types of Dunes**:
    - **Barchans** - These **crescent-shaped dunes** have their **points or wings oriented away from the wind direction.** They form in areas where wind direction is consistent and moderate, and the sand moves across a relatively uniform surface.
    - **Parabolic Dunes** - Resembling **reversed barchans,** parabolic dunes develop on partially vegetated sandy surfaces. The **wind direction remains the same** as for barchans, but the **presence of vegetation influences their shape.**
    - **Seif Dunes** - Seif dunes resemble barchans but have **only one wing or point.** They form in areas where wind conditions shift, causing the single wing of a seif to extend significantly in length and height.
    - **Longitudinal Dunes** - These dunes appear as **long ridges aligned with the direction of prevailing winds.** They form where the sand supply is limited and wind direction remains constant. Longitudinal dunes are typically elongated and low in height.
    - **Transverse Dunes** - Aligned **perpendicular to the wind direction,** transverse dunes form when there is a consistent wind direction and a sand source that extends at right angles to the wind. These dunes can be very long but generally low in height.



Ripples:

- Ripples are small, regular, wavelike undulations that form **perpendicular to the direction of the prevailing wind.** These features are created as wind moves across loose sediment, arranging it into distinct, patterned shapes.

Loess:

- Loess is a type of wind-deposited sediment consisting of fine, **fertile dust.** It is **yellowish, friable, and highly porous, allowing water to sink in quickly, which often results in a dry surface.** Loess can form extensive deposits, as seen in the **Loess Plateau of northwest China,** which covers about 250,000 square miles and reaches depths of 200 to 500 feet. Known locally as **'Huangtu'** (yellow earth), loess deposits in China are similar to those in Germany, France, Belgium, and parts of the Midwestern USA. In these regions, loess was deposited from material at the edges of ice sheets during the Ice Ages and is sometimes called adobe.

===== XXXXX =====

(1) Wind
(2) River
(3) underground Water
(4) Wave
(5) Glacier

Shape Landform ← Geomorphic Process ← (1)–(5)

Geomorphic Process → Erosion, Deposition

Land form by River

Erosion

(1) Canyon
(2) Gorge
V-shaped
(3) Pothole
(4) Plunge Pool & waterfall

Deposition

1. Delta
2. Flood Plain
3. Alluvial Plain
4. Riverline Island
   ↳ Majuli in Brahmaputra
5. Natural levee

Special laform

Meander & oxbow lake

2. River Terrace